



source code

CraftBot: Physical Prototyping Through Large Language Models

CraftBot is a harness for architect AI agents. It is agent-agnostic and can be used with any modern multi-modal LLM (Large Language Model). CraftBot reads a design brief, grounds itself in domain knowledge by ingesting documents, images and other references, and outputs Python code that defines a building procedurally. Instead of producing meshes or images, CraftBot writes scripts that construct architectural geometry in Blender. From there it can produce the industry's standard representations: floor plans, sections, elevations, BIM models, bills of quantities etc. It can run in a fully automated loop of code generation, execution, visual feedback and revision. The main research question is whether LLM agents can participate in architectural design when constrained to work through executable CAD code.

The project is documented through a GitHub repository holding code, iteration steps, references and outputs of the experiments done so far, enabling anyone to build on top of CraftBot. Construction manuals are included as extracted text summaries with links to the originals, which are left out for copyright reasons. Python and Blender were chosen for ease of access: both are open source, maintained, widely used, and free for commercial use.

Link to the repository: <https://github.com/lukapiskorec/craftbot>
You can view and manipulate the outputs of the previous CraftBot experiments using the online interface: <https://lukapiskorec.github.io/craftbot>

Proof of concept: experiments

CraftBot was developed and tested through a series of fourteen experiments conducted between November 2025 and August 2026. Each experiment starts with a short design brief and reference files. The early experiments work from pictures alone: a carport with a king post truss, the same carport rebuilt with a gothic hammer beam truss, a small steel shelter, Jean Prouvé's demountable 6x6 house. The later ones add text sourced from reference manuals. Experiment 08 was given a 27-page special issue of The Architects' Journal from 1986 describing the Segal method of timber self-build, and produced a house in that method. Experiment 09 read a guide to cross-laminated timber and built a ten-storey block, while experiment 11 started from a Canadian manual on wood-frame house construction to produce a model of a hip roof.

After the input resources are read, the user-defined brief is interpreted and expanded into a building concept and subsequently modelled in Blender using Python code as procedural instructions. Then the agent uses the code to generate the 3D model of the building and takes screenshots of it using headless Blender: no Blender window is actually opened, the agent is able to run the whole rendering pipeline through a command line by itself. The screenshots are then evaluated by the agent and any deviations from the brief or mistakes in the model noted. The agent then goes back to refine the Python code, saving each version of the model as a design iteration until it converges with the brief.

In the first version of CraftBot (ChatGPT 5.1 as agent), this iterative improvement loop was manual and required human attention to execute. The new version with Fable as the agent is able to execute this loop in a fully autonomous manner. Although no human input is needed in principle, future experiments will test a more hybrid approach. Human input might help better guide the convergence of the model with the brief, especially in the later iterations of the design.

As of September 2026, the experiments are still ongoing and can be found on the project's GitHub page.

Agent team

The most recent iteration of CraftBot (September 2026) implements a team of agents, each with its defined role, inputs, working procedures, and memory:

- **CraftBot**, orchestrator, writes down the brief as it understood it, decides what is in scope, compiles the design rationale document at the end and reports back to the user.
- **Designer** produces the spatial and construction concept and turns it into a numbered checklist of requirements, each one with its source and a way to test it.
- **Researcher** searches the manuals for what the concept needs, picks the chapters that apply, reads them and crops out the figures the Designer needs.
- **Builder** implements the concept as versioned Blender Python scripts, renders them and runs the automated checks for geometry overlap, floating elements etc.
- **Inspector** looks at the renders of the model, which it compares against the requirements and the reference materials, and reports what is missing, misplaced or unlike the reference.
- **Runner** works in the background, exporting each finished version to the web viewer and archiving the run.

The rule that holds the team together: every hand-off is a file on disk, never a chat message. Hand-off documents include the brief, concept, sources, requirements, design notes, version notes, inspection reports etc. Any agent can be terminated and restarted from those files, and the human reading afterwards gets the same record the agents worked from. To aid their work, agents use skills, documents that describe specially designed procedures to accomplish specific tasks, placed in the skills/ folder of the project repository. To ground their work in architectural domain knowledge, they consult expert manuals and guide documents, while reusable Python scripting tools are available in the tools/ folder. These materials and the resource structure form the backbone of the agent harness.

The pages that follow reproduce, unedited, the hand-off documents of this run: the brief as CraftBot understood it, the Designer's concept, the design notes and the Builder's version notes.

Brief as understood

This single-variation run builds a square room twice as tall as its clear width, with one entrance and daylight from the roof or openings close to it. Thin timber slats form the enclosure and the exterior structure. The red building photographs guide construction character and scale; the other two images guide a series of undulating trusses that give the outside an expressive profile while preserving a simple interior. The 2:1 proportion applies to the clear room, with the exterior trusses allowed to project beyond it. No reference feature is deliberately dropped at this stage; dimensions inferred from photographs must be identified as estimates.

Inputs and assumptions

- Input images: 01-soane-front.jpg, 02-soane-outside.jpg, 03-soaneinterior-1.jpg, Erne_Dachbau_778.webp, Tchumi-02.jpg.
- No input image is overridden at setup. The user's specified square plan, 2:1 clear-room ratio, one entrance, thin slats, and quiet interior control the synthesis.
- One building, no alternative variations requested.
- Timber structural and enclosure members must remain visibly thin slats, including built-up truss members where needed. Connection adequacy and weatherproofing must be stated honestly; do not silently replace the slat system with conventional solid heavy framing.
- Existing prompt files for other models are not instructions for this run.

Concept-stage scope decisions

The adopted clear dimensions are 3.00 x 3.00 x 6.00 m. A provisional 3.60 x 3.60 x 7.20 m scheme was reduced after the photo calibration suggested about 2.8 m exterior width with substantial perspective uncertainty. The adopted width is a design estimate close to that source; the exact 2:1 height follows the user even where the reference appears less tall. Seven truss frames are spaced at 0.50 m.

Open high clerestories provide daylight without glazing. This is a conceptual timber pavilion enclosure; weatherproofing is not established. Ground support, fixings, member capacities, and hold-down design remain unverified. These performance limits do not waive geometric bearing, contact, bracing, or visual inspection requirements. A usable 0.90 x 2.10 m entrance deliberately replaces the unusually low reference aperture. The second and third structural images guide timber geometry; their glazing and unrelated enclosure materials are not copied.

Experiment 16 concept — GPT-6

Concept-stage release for v01, 2026-09-14, with the recorded sole-runner clarification. The brief controls the synthesis; all measurements below are metres unless mm is written. Values marked **mine** are design assumptions, not measured or engineered facts. This specification preceded geometry; the built v02 review is recorded in design_notes.md on 2026-09-15.

Spatial concept

One square, empty room, with red vertical timber slats enclosing a quiet shaft. Clear inner faces are x/y = +/-1.500; finished floor z = 0; flat ceiling underside z = 6.000. Thus clear width and depth are 3.000 and clear height is exactly twice either dimension. These are **mine**, selected as a compact room scale consistent with the red references, not a surveyed size. No intermediate floor, stair, gallery or furniture is required. One centered entrance faces negative y; clear opening 0.900 wide by 2.100 high (**mine**, normal passage replacing the unusually low aperture in the interior photograph). An open doorway is the one entrance; a movable leaf is not required by the brief.

Daylight enters a 0.750-high open clerestory band on the two x walls, z = 5.250 to 6.000, immediately below the level ceiling (**mine**). Timber posts and rails cross that band but no opaque infill does. North and south walls remain boarded to the ceiling, echoing the high central red wall in the interior reference. The opening has no glazing: the all-slat brief is interpreted as an open-air architectural pavilion, with rain-tightness and conditioned occupation outside this geometric model. This assumption must remain visible in the report. The lower wall boarding and roof cover are geometrically closed, so intentional daylight openings remain concentrated above.

Construction concept — research questions cite this section

Seven parallel transverse truss frames span x and repeat along y; their nominal stations are -1.500, -1.000, -0.500, 0, 0.500, 1.000, 1.500 (**mine**, six bays at 0.500). Side-wall inner chord lines remain vertical outside the square enclosure. Each frame combines triangulated deep side columns with a triangulated roof span. The outer chord varies by height and frame station, so a family of straight-stock polygonal trusses makes a waving exterior around the straight interior. End-wall boarding is held by light rails and jamb assemblies tied into the first and last transverse frames. The two side walls carry the roof; the entrance wall does not require a roof transfer across its doorway.

Released catalogue: enclosure, floor and ceiling slats B = 24 x 90 mm; primary chords and diagonals S = 36 x 90 mm. Primary chords use two separately legible S slats straddling one S web plane; timber packing/node straps use the same stock. No solid glulam, plywood

gussets, thick posts or sheet cladding. Slat sizes are **mine**; the Researcher's scope verdict finds useful thin-member precedents but no capacity certification. Thin paired chords are connection analogies, not a nailed-truss design taken from a manual. Ripped edge slats and packers may be smaller than the catalogue but never larger in either cross-section dimension.

Ground to roof: a flat grade datum below the floor; built-up crossed slat sleeper mat on grade; close-laid floor slats; column inner and outer feet on continuous slat sole runners; column webs delivering roof reactions and wind shear to those feet; roof lower chords seated on paired side-column heads; triangulated webs to faceted upper chords; short roof slats spanning between adjacent frame top chords. Level ceiling slats are face-fixed to roof lower chords and establish the simple 6.000 m room datum. The ceiling is held by concealed fixings, not falsely described as bearing on supports above it. All splice positions land on nodes or supports. Concealed fixings, hold-downs and damp separation remain connection/performance design requirements, not invented timber capacity.

Lateral concept: triangulation of each x-z side/roof frame handles transverse action; longitudinal y-z bracing in the exterior side-wall depth connects the seven frames at several heights; explicit timber diagonals at roof and floor levels collect shear to these braced lines. A boarding layer alone is not claimed as a diaphragm. Both longitudinal end bays are braced; entrance remains free. Whole-building uplift, overturning, foot anchorage and connections require engineering beyond the manuals.

Photo rule set

P-01 — 01-soane-front.jpg gives overall topology: a tall compact red enclosure sits inside exposed slender timber framing. The visible main face has five prominent paired vertical frame lines/four bays; the new seven-frame depth rhythm is a deliberate synthesis with the roof references, not a count copied from that face.

P-02 — 02-soane-outside.jpg gives construction character: paired thin members, open gaps, diagonal braces, and modest short horizontal projections. Preserve that assembled, slender reading; do not substitute chunky solid posts.

P-03 — 03-soaneinterior-1.jpg gives spatial character: uninterrupted tall red vertical boards, a level timber floor, one centered low aperture, and light concentrated at the top. Preserve the room and high-light hierarchy; normalize the opening to a usable entrance. Board end joints may stagger but wall planes remain flat.

P-04 — Erne_Dachbau_778.webp gives roof articulation, not room size: changing-depth trusses with triangulated webs and segmented chord direction. Model actual web triangles and changing roof depth; a wave-shaped solid panel is insufficient.

P-05 — Tchumi-02.jpg gives an outer-edge rule: repeated slender straight pieces form rising and falling profiles, with projected ends creating depth and shadow. Apply this to the frame series and outward side-column profiles; its glass and dense facade screening are not copied.

P-06 — Calibration is approximate. In the 933 x 1400 front photo, the outer stair rises about 280 image pixels over approximately 10 risers. Assuming 0.170 m/riser gives 165 px/m near that stair plane. The principal face spans about 470 px and the red-plus-upper enclosure about 680 px, suggesting roughly 2.8 m width and 4.1 m enclosure height under that local scale. Perspective, stair/face depth separation and obscured levels make those figures low confidence (at least +/-25%); they cannot establish clear room size or override the brief's exact 2:1 ratio. Adopted 3.000/6.000 dimensions are programme decisions. The low interior aperture is not used as a standard-door calibration.

P-07 — Colour follows the red-building references: muted oxide-red boarding and selected web slats, natural pale timber for paired chords and roof/ground slats. Colour is a finish on timber, not another construction material.

Questions settled by the research verdict below

Q1: Which manuals justify narrow solid-timber floor and roof boarding at 600 mm support centers; minimum thickness and end-bearing/joint rules?

Q2: What manual truss member sizes, clear spans, node/gusset and inter-truss bracing figures offer a valid precedent; which parts do not establish capacity for the proposed 36 x 90 paired-slat system?

Q3: Does any manual cover a 6.0 m tall laterally braced slat truss column; what limits must explicitly be recorded, and what bearing/opening/base-detail rules can be transferred without claiming adequacy?

Fixed geometry and construction numbers for the Builder

All values in this section are **mine**, unless the source table below says otherwise. Profiles are centerlines; faces must be offset by actual member sections and clipped at joints. No part may invade the clear 3.000 x 3.000 x 6.000 room other than the plane of its enclosure. Assembly tolerances may close joints but may not change headline room dimensions.

Symbol / family. Fixed value and datum. Purpose.

ROOM_W / ROOM_H. 3.000 / 6.000, inner finished faces. Exact user ratio.

FRAME_Y. -1.500 + 0.500 j, integer j=0..6. Seven frames.

B. thickness 0.024, visible face width 0.090. Wall, floor, ceiling and roof slats; final course ripped to boundary.

S. thickness 0.036, in-plane face width 0.090. Chords, webs, rails, ledgers, braces and sleepers.

WALL_FACES. inner x/y = +/-1.500; outer faces = +/-1.524. Flat vertical B boarding; outer structure never moves these faces.

COLUMN_INNER_X. +/-1.600, centerlines. Nearest chord face +/-1.555 leaves room for wall attachment battens.

COLUMN_Z. -0.024, 1.500, 3.000, 4.500, 6.024. Five nodes, four column panels.

COLUMN_DEPTH. d(0,j)=0.300; d(4,j)=0.600; intermediate d(k,j)=0.300+0.300 sin²(pi*k/4+pi*j/6). Outer chord x = +/- (1.600+d); visible lateral wave; depth 0.300.0.600.

CHORD_LAYERS. paired S centerplanes y=FRAME_Y +/-0.036, thickness 0.036; web plane y=FRAME_Y, thickness 0.036. Faces meet at y=FRAME_Y +/-0.018; total built-up thickness 0.108; each chord remains visibly a thin slat.

ROOF_X. -2.200, -1.600, -0.800, 0, 0.800, 1.600, 2.200. Seven roof panel stations, supports exactly over inner and outer column heads.

ROOF_BOTTOM. lower chord centerline z=6.069, underside z=6.024. Direct 0.090-wide seat on column heads; ceiling top z=6.024.

ROOF_TOP. upper chord node z=6.650+0.300 sin²(pi (x+2.200)/4.400)+0.200 sin²(2*pi*j/6). *Faceted changing-depth roof; approximate upper centerline range 6.477..7.123; seven ribs sampling three offset levels through a rise and fall, same flat inner ceiling.*

ROOF_COVER. B slats run generally along y between adjacent frame stations; ends land on top chords. Additive x/y height field gives planar quadrilateral facets, allowing straight stock; joints shared at support stations.

FLOOR_STACK. B z=-0.024..0; upper sleepers z=-0.060..-0.024; lower sleepers z=-0.096..-0.060; flat grade z=-0.096. Two crossed thin-slat bearing layers under the floor and column feet.

GROUND_MAT. outer envelope 4.600 x 3.220, centered; upper transverse S sleepers at each FRAME_Y; lower longitudinal S sleepers at <=0.500 spacing plus all column-foot lines. Ground-level load distribution; no cantilevered thin sole beneath a column.

SOLE_RUNNERS. continuous longitudinal S runners at inner/outer column-foot lines, integrated in upper layer

z=-0.060..-0.024; upper transverse sleepers stop against runner faces. Feet remain at z=-0.024; lower longitudinal sleepers directly beneath each runner support every foot footprint; no third layer or overlapping cross-joint.

WALL_RAILS. S horizontal attachment rails at vertical spacing <=0.750; add at z=0, 2.100, 5.250, 6.000 as applicable. End joints supported; trim rails clear of entrance and clerestory; side battens fill the 0.031 gap to inner chord faces.

WALL_STOCK. B board joints at rail heights, maximum individual length 3.000. Stagger adjacent board joints; no unsupported butt joint.

ENTRANCE. x=-0.450..+0.450, y negative wall, z=0..2.100. Clear opening; doubled S jambs and S header side plies with >=0.090 seat each end; red board infill above carried through header to jambs.

BRACING. longitudinal braces in both end bays of both side walls, between column node levels; S rails link all seven frames at node levels. y-direction shear carried to ground without relying on skin.

PLAN_BRACING. S diagonal pairs in roof and ground-mat planes, crossing in separate thickness layers. Collect shear across 3.000 m room; transverse x-z frames carry x-direction action.

SPLICES / NODE_STRAPS. paired S splice straps nominal length 0.360, centered on a supported splice; all member ends on shared node planes. Visible timber contact; exact nail or screw schedule is unverified; do not substitute plywood gussets.

Use a single alternating diagonal per roof or column panel plus posts/ties at panel boundaries, so each panel is a genuine triangle system rather than decorative fins. At crossing braces use separate faces, not overlapping volumes. Chord-to-web connections are layered face contacts; chord-to-chord corners use complementary mitres and, where needed, paired S straps. Roof lower and upper chords are continuous or spliced at web nodes. Roof boarding meets the actual faceted upper-chord bearing plane; do not raise boards until they float above supports. An exact joint may require a short S blocking/cleat derived from the stated member faces; this is a connection detail within the catalogue, not authority to add heavy members.

Base assembly clarification for v01: the continuous sole runners form part of the upper crossed-mat layer, not an extra layer above it. Each foot bears on the runner top at z=-0.024; that runner bears on the lower sleeper top at z=-0.060, and the lower sleeper bears on grade at z=-0.096. Upper transverse sleeper segments butt to the longitudinal runner faces with concealed connection design unverified. Maintain full foot bearing within the stated thin-slat catalogue and floor supports at <=0.500 m; no unsupported runner projection carries a column. All base-detail dimensions here are **mine**.

Side walls use flat red B slats to z=5.250 and open clerestories above. End walls use flat red B slats to z=6.000 except the entrance. Floor and ceiling B slats run along y, with ends on the 0.500 frame grid. Ceiling fixings carry its gravity load into lower chords. Roof B slats fully cover their facets; edge seams are allowed, open decorative gaps through the roof are not. No construction claim of rain-tightness is made for board seams or valleys.

Roof edge boarding extends to the outer faces of the edge chords, at most 0.045 beyond the outer x node and 0.054 beyond the first/last y frame. This minor extension closes the cover over the frame faces, not a new cantilever. Wall corners use complementary cuts or one wall stops at the other's inner face; double thickness at a corner must not reduce clear dimensions. Bearing at column heads and entrance-header ends is 0.090, a **mine** timber seat, not an application of the manual's masonry bearing minimum. Roof board ends split on frame centerplanes and bear on the relevant 0.036-wide chord ply; connection and splitting adequacy are unverified.

Source-to-rule-to-number verdict before geometry

Source. Rule taken. Model number / disposition. Verdict.

User brief; P-06 photographic scale. Square clear room; clear height twice width. ROOM_W=3.000; ROOM_H=6.000. Brief controls ratio; width is a photo-informed estimate.

P-01/P-02 red-building photos. Thin paired framing and flat red vertical enclosure. S=36 x 90 mm paired; B=24 x 90 mm. Sizes mine; construction character retained.

CMHC Table 22, p.287; Table 35, p.305. Lumber-board thickness depends on supports. Source minimum 19 mm at600 mm; adopted B24 mm at500 mm. Useful thickness precedent; no certification of narrow boards or diaphragm.

FRIM Table1 / Appendix I Sheet1. Drawings and schedule govern conflicting floor-board prose. Source22 x145 mm at610 mm supports; adopted24 x90 mm at500 mm. Narrower slats are brief-driven, mine; source30 mm prose rejected.

FRIM Appendix I Sheet9, PDF p.58. Actual web triangles, upper/lower chords, slender stock. Source35 x72 mm over5.770 m, rise1.195 m; adopted S36 x90 with paired chords,3.200 m inner support span /4.400 m outer span. Precedent only; no scaling or capacity claim.

FRIM Sheet9 roof plan; CMHC Ch.11 permanent truss bracing. Connect trusses with longitudinal and diagonal restraint. Source22 x97 mm braces at1.220 m trusses; adopted S braces tying500 mm frames at column levels and roof/floor planes. Bracing topology transferred; sizes and spacing mine.

FRIM Sheet9 gusset detail. Nodes need designed fasteners and force transfer. Source9 mm plywood each side with3.3 x63 mm nails; spacing46/23 mm and edge16 mm. Not implemented: plywood conflicts with all-slat brief; paired S node straps and concealed connections require engineering.

CMHC Table25, p.290. Prescriptive stud cases have limited unsupported height. Roof-only exterior case3.0 m; table maxima3.6 m exterior and4.2 m interior; adopted6.0 m truss-column with1.5 m panels. Not covered: no claim of stud-table or truss-column adequacy.

CMHC Fig.68, p.109. Opening infill transfers through header to jambs. Clear entrance0.900 x2.100 m; doubled S jambs,0.090 header seats. Opening grammar transferred; member/connection sizes mine.

CMHC Fig.86, p.125. Roof reactions require real seats and aligned supports. Inner/outer column heads align x=+/-1.600,+/-2.200 with roof nodes;0.090 seats. Bearing logic transferred; roof is a truss, not a rafter/collar-tie prescription.

FRIM Sheet7 base; CMHC Fig.47, p.79. Bases require support, anchorage and moisture separation. Model crossed slat mat0.096 m under floor; no invented steel shoe/concrete footing. Ground contact modeled; anchors, uplift, moisture and soil capacity not verified.

Erne and Tchumi images, P-04/P-05. Segmented triangulated depth and repeating wave. Seven ribs; fixed COLUMN_DEPTH and ROOF_TOP formulas above. Geometric synthesis, no literal industrial dimensions transferred.

Builder figure set (shared provenance retained): ../references/manual_frim_sheet9_p58_truss_elevation_heel_span.png, ../references/manual_frim_sheet9_p58_roof_bracing_plan.png, ../references/manual_frim_sheet9_p58_truss_gusset_nailing.png, ../references/manual_cmhc_p109_fig68_wall_opening_lintel_jack_cripp le.png, ../references/manual_cmhc_p125_fig86_rafter_heel_loadbearing_wall.png, and ../references/manual_cmhc_p079_fig47_wood_sleeper_on_slab.png. The gusset and base images explain what is not copied, as well as the connection principles.

Sections are plausible visual propositions for a thin-slat pavilion, not calculated. Whole-building strength, buckling, joint forces, fastener spacing, uplift/overturning anchorage, soil bearing, fire, durability and weatherproofing are not verified.

Adopted dimensions against photo estimates

Quantity. Reference estimate. Adopted. Verdict.

Compact enclosure width. about 2.8 m, at least +/-25% uncertainty. 3.000 m clear. +7% before unknown wall thickness; reasonable room-scale inference.

→ spawns → reports back

The CraftBot agent team, September 2026: who spawns whom, and what each hands back.

Tall enclosure height, about 4.1 m in front projection, at least +/-25% uncertainty. 6.000 m clear. Deliberate brief override to achieve exact 2:1.

Prominent front frame rhythm, five lines / four bays, seven transverse frames / six depth bays. Synthesis with expressive repeated roof; not a photographic replica.

Low aperture, no reliable absolute scale. 0.900 x 2.100 m clear. Programme assumption for a usable entrance.

Raised floor and exterior stair, present in reference, floor 0.096 m above grade; no stair. Single room brief does not require raised tower access; simple low threshold.

Experiment 16 design notes — GPT-6

2026-09-14 — v01 concept: two envelopes

The five images share a useful distinction: a calm enclosure can sit within a much more active external assembly. Adopt a flat red square room and put changing depth in exterior column trusses and roof trusses. Rejected a wavy inner wall/ceiling because the brief expressly asks for the inside to remain simple. Rejected separate decorative fins because the exterior must read as a series of trusses with actual web triangles and identifiable load paths.

2026-09-14 — v01 concept: scale and photo calibration

The front-photo stair supplies a weak scale anchor: about10 assumed170mm risers across280 pixels gives about165px/m at the stair plane. That suggests roughly2.8m facade width; it does not measure a clear room or cancel perspective. Initial3.6m width would have added about29% to that estimate. After scope review, adopted3.0m width (+7%, within the stated uncertainty) and exactly6.0m clear height. Rejected a claim of surveyed dimensions. The height is a deliberate brief-led increase over the approximate4.1m projected photo enclosure. The user ratio is the controlling dimension.

The reference's low opening is not credible as a normal-door calibration. A0.9x2.1m opening provides one usable entrance. Raised floor and exterior staircase from the red tower are not copied: a0.096m slat-mat floor keeps the requested single room direct and compact. No second entrance, movable door leaf, stair or gallery has been added.

2026-09-14 — v01 concept: thin slats and load paths

The repeated catalogue is24x90mm boarding and36x90mm structural slats. Paired chords bracket a single web layer; deep truss action comes from separated chords, not a solid thick beam. Roof reactions go to the paired side-column heads, down triangulated inner/outer chords to sole runners and a crossed slat mat. Longitudinal braces tie frame stations to ground; explicit floor/roof diagonals collect lateral action. Rejected conventional glulam posts, plywood gussets and steel brackets as modeled elements because they would undermine the entirely-thin-timber brief. Concealed fasteners and anchors are acknowledged as unresolved construction necessities.

The manuals establish only analogies: CMHC19mm boards at600mm support spacing and FRIM35x72mm truss members over5.77m. Our24mm boards at500mm spacing and36x90 paired chords are mine, not manual-certified. The6m braced columns, connections, base anchors and soil interface have no matching prescriptive design. No structural adequacy claim is made.

2026-09-14 — v01 concept: fixed faceted expression

Seven frames at500mm spacing carry an additive roof height field: a low curved profile sampled across x plus a changing vertical offset along y. Each interval remains a planar facet, so straight short slats can close the roof and bear at their ends. The outer side-column nodes also vary in depth while their inner chords stay vertical. Rejected an arbitrary double-curved skin that would force twisted long boards; rejected a simple pitched roof repeated unchanged because the two expressive references ask for stronger articulation.

All characteristic profiles and paired chord layer offsets are fixed in concept.md before geometry. A Builder may fit and clip joints within those profiles; changing the room, number of ribs, exterior expression or member catalogue is a design question.

2026-09-14 — v01 concept: enclosure and daylight scope

Opaque lower walls concentrate light at0.75m-high clerestory bands immediately under the6m ceiling on both side walls. End walls remain full-height red boards, as in the interior photo. The roof is fully covered by slats and ceiling is a level slat plane. No glass or membrane is modeled. CraftBot confirmed the open-air pavilion reading on2026-09-14: this is a geometric architectural proposal with no demonstrated rain-tightness or conditioned-building performance. Rejected adding glazing merely to make the model appear weatherproof.

Deviation ledger at concept release

Category: Decision. Reason / rejected alternative.

Source deviation. 3.0x3.0x6.0m clear volume. Photo width estimate retained approximately; exact user2:1 ratio overrides projected photo height.

Source deviation. Seven transverse frames rather than copying five prominent front lines. Series synthesizes roof references; red photos guide scale and assembly, not a literal replica.

Source deviation. Usable0.9x2.1m entrance; floor close to grade. No requirement for low aperture or raised tower stair.

Source deviation. All-slat paired nodes instead of FRIM plywood gussets and steel base shoes. User material palette; connection capacity not inferred from substituted detail.

Considered, kept. Red vertical enclosure, natural chords, triangulated webs, high side light. Features agree across the red-building and expressive references.

Considered, kept. Level inner ceiling despite varying roof trusses. Expressive exterior must not disturb simple interior.

Not verified. Strength, buckling, fasteners, uplift/overturning, soil bearing, moisture, fire, weatherproofing. Outside source coverage and geometric checks; no construction certification.

Phase-2 comparison and structural review

Performed on v02 on 2026-09-15. The comparison and structural-review records below supersede the concept-stage placeholder. The original concept decisions and v01 clarification remain as the design history.

2026-09-14 — v01 build clarification: sole-runner datum

Accepted the Builder's integrated upper-layer sole runner assembly. Continuous longitudinal S runners occupy z=−0.060..−0.024, and upper transverse sleeper segments stop at their faces. Column feet remain at z=−0.024. Lower longitudinal sleepers sit directly beneath the runners and every foot footprint, carrying the load from foot to runner to lower sleeper to grade without a thin-slat cantilever. Floor support spacing remains ≤=0.500 m; fitted butt contacts replace intersecting volumes, with concealed connection design still unverified. This clarifies R-16 and the SOLE_RUNNERS row without changing stock or clear room dimensions.

Rejected placing a third 36 mm sole layer above the mat because it would raise feet to z=+0.012 and contradict the released COLUMN_Z datum. Rejected omitting the runner because a continuous bearing and tying line is part of the stated base assembly. The accepted arrangement retains that line within the existing 96 mm

ground-to-floor stack; the crossing mat is continuous through the lower layer where upper transverse pieces are interrupted.

2026-09-15 — v02 independent structural review

This review concerns the built evidence in version notes.md v01/v02, inspection_v01.md, inspection_v02.md and inspection_v02_room.md. The Designer did not read the geometry scripts or renders, and did not infer compliance merely from the concept. The Builder supplies numerical and fitted-plane evidence; the room Inspector supplies the separate visual channel. The reference images were set aside for this pass.

Element family / condition. Built load-path argument and evidence. Finding / action.

Roof slats. Version notes report short boards spanning adjacent 0.500 m frame stations with undersides on the actual beveled upper-chord planes and fitted facet boundaries. Inspector views 11/16 show complete coverage and support lines. Gravity goes through those contacts to the upper chords. Retain R-15. No reported unsupported board or visible gap. Every contact area is not independently measurable in renders; no roof-board capacity or fastening claim.

Roof trusses. Notes identify upper/lower chords, web nodes, column-head seats and complementary joints; views 08/11/12 confirm web triangles and supported heels. Reactions pass through the web/chord assembly to the paired inner and outer column heads. Retain R-08/R-17/R-22. The 90 mm seats are Builder evidence, visually consistent at view 12. Connection forces and the adequacy of segmented chord splices remain unassigned.

Ceiling. Notes explicitly put ceiling top at z = 6.024 against the lower-chord underside. Its gravity load goes through concealed face fixings into the lower chords, then to column heads. View 19 confirms a level unobstructed plane. Retain R-23. This is a proposed fixing-dependent load path, not bearing from above. Fixing number, withdrawal resistance and fire performance are unverified.

Side and end-wall boards / rails. Notes report rails at no more than 0.750 m, supported staggered board joints, ripped side attachment battens and end rails terminating at inner-chord faces. Boards transfer gravity and out-of-plane loads through their fixings and rails to exterior chords; lower board ends also meet the floor edge where detailed. Views 06/07/10/19 confirm the enclosure and upper openings. Retain R-05/R-21. Exact rail spacing and every concealed attachment depend on Builder evidence; no wall shear capacity is assigned to the boarding.

Entrance discontinuity. Notes name a header with 90 mm seats, doubled jamb plies, a ripped edge slat supporting both plies, and removed infill/rails inside the passage. The header carries infill above into the jambs, then into the supported mat edge. The transverse roof frames avoid using this opening as their principal roof support. View 15 confirms the header/jamb grammar and clear passage. Retain R-02/R-18. No uncarried end or obstructing member is reported. Header, jamb and connection capacities remain uncalculated.

Truss-column feet and soles. v02 actual-mesh assertion checks all 56 horizontal foot faces against both sole and lower sleeper, including support top levels and full 90,000 mm seat width. The sole is at z = −0.060..−0.024, the aligned lower sleeper at z = −0.096..−0.060, then grade. View 14 shows seated feet. R-16 is resolved geometrically. The former maximum 0.753 mm unsupported edge per side was removed by fitted foot-end bevels, not hidden by a relaxed tolerance. No further strips or larger stock justified.

Floor and ground mat. Notes report floor boards spanning the 0.500 m stations, upper transverse sleeper pieces fitted against continuous longitudinal sole runners, and lower sleepers directly below column-foot lines. Floor gravity passes to upper sleepers/runners, then the lower layer, then grade. Views 09/18 show continuous support and no isolated floor strip. Retain R-14/R-16. No modeled retaining wall, slope or suspended undercroft requires another foundation detail. Soil pressure, settlement and long-term timber-to-grade exposure are unverified.

Transverse x direction. Each x-z roof/column frame contains actual triangles. Roof-plane braces collect lateral action over the room width to the exterior truss-column lines; the frame chord/web forces reach their separated feet and the mat. Frame view 08 and roof-bracing view 17 confirm the relevant members. Retain R-08/R-20. This is a braced truss proposal, not a moment frame. Global resistance still depends on designed node connections and ground hold-downs; contact alone cannot carry tension or overturning.

Longitudinal y direction. Built notes identify longitudinal ties at column nodes and diagonal end bays connecting the seven frames. The nominal 3.0 m frame array has 0.500 m bays. Roof/ground-plan diagonals collect shear into these braced side lines, whose braces continue to the mat. Views 07/08/17/18 confirm the members. Retain R-19/R-20. Fitted tie/brace ends were segmented to avoid intersections in v01; their force continuity requires concealed fasteners, which are named but not designed. Skin is not claimed as the diaphragm.

Member slenderness and restraint. The reported system remains 36 x 90 mm structural slats, paired for chords, with roughly 1.5 m column panels and 0.500 m frame repetition. A single 36 mm ply has a weak-axis radius of gyration of 36/sqrt(12) = 10.4 mm; if a 1.5 m panel were an effective buckling length, L/r would be about 144. This is the Designer's dimensional indicator, not a stability calculation; paired-ply composite action and effective restraint are not established. Keep the explicit unsupported-engineering finding in R-26/R-28. The slender column system needs member, restraint and connection design before construction. No evidence supports an arbitrary larger post or a change to the thin-slat brief during this geometric review.

Sliding, uplift, moisture and drainage. Notes do not model a ground anchor, moisture break or weatherproof membrane. The roof is faceted and fully covered with slats; upper clerestories are intentionally open. A seated foot transmits compression but does not establish a tensile ground connection. Roof seams/falls have no demonstrated waterproof drainage detail. Record as unverified under R-28, not as proven stability or weather protection. No glazed enclosure, steel shoe, concrete foundation or membrane is silently added. This remains the accepted conceptual pavilion scope.

The two reported geometric checks are 0 penetrating pairs above 1 mm and 0 floating members at a 2 mm contact tolerance, with no exclusions. They neither prove global connectivity to ground nor adequate bearing by themselves. The separate v02 full-foot assertion uses actual horizontal mesh faces and a 1 micrometre precision tolerance; it supplies the specific bearing evidence missing from v01. Roof and doorway seats use Builder dimension/assembly evidence with visual consistency, not an equally comprehensive independent mesh-bearing audit.

No new geometric defect is established by this review. The explicit lateral members, full-foot correction and documented opening/support paths meet the geometric requirements without another version. This conclusion is limited to the supplied built records and Inspector findings. It does not certify that the pavilion is stable, anchored, weatherproof or ready to construct. R-14/R-15/R-21/R-22/R-23 retain their stated partial-visibility limits; those limitations are not waived visible defects.

Rejected unnecessary changes: added base strips would duplicate the resolved foot bearing; a third sole layer would change the accepted datum; oversized posts or plywood gussets would replace the user's thin-slat system without an engineering design; glazing or a membrane would expand the accepted pavilion scope. None is introduced as a speculative fix.

2026-09-15 — v02 source-versus-model comparison and final design disposition

The mechanical comparison Inspector opened all five original input photographs, the six concept-cited manual snippets, and v02 views 01–04, 06–08 and 10–19. Its complete row-by-row record is inspection_v02_comparison.md. The following Designer table accepts or acts on those actual-image findings; it is not a comparison inferred from the concept text. No additional source search was

undertaken.

Reference image / figure. In the reference. In v02, from comparison Inspector. Change or retained with reason.

01-soane-front.jpg. Tall compact red volume in thin framing; five prominent front lines, raised floor and exterior stair. Views 03/06 show the red volume, pale/red frame and centered entry; the seven-frame rhythm is along depth. Retain. A 3.0 m room width is close to the approximate 2.8 m photograph estimate. The 6.0 m height deliberately overrides the approximate 4.1 m photo height to meet the user's exact 2:1 ratio. Seven depth frames and a low threshold are recorded synthesis choices.

02-soane-outside.jpg. Paired slender timber, diagonal braces, open gaps and small projections around red boarding. Views 01–04/07/08 show paired pale members, red webs, repeated projections and open structural depth. Retain. This is the strongest direct construction-character transfer. Projections are not literal survey dimensions; no heavy post substitute appears.

03-soaneinterior-1.jpg. Tall red vertical boards, high side light, simple floor and unusually low aperture. Views 06/10/19 show the tall clear room, level ceiling and upper side openings with the normal entrance. Retain. Usable 0.9 x 2.1 m entry and a flat ceiling preserve the programme. View 19 hides end boards to expose construction, so the enclosure comparison also uses view 10 and front view 06; it is not a photometric daylight test.

Erne_Dachbau_778.webp. Repeated segmented trusses with genuine triangles and changing depth. Views 01–04/08/11/16 show a compact series of triangulated roof ribs, changing profile and slat cover. Retain. The truss system is adapted to the pavilion's scale. Its industrial dimensions and joint capacities are not transferred.

Tchumi-02.jpg. Repeated straight members describe a rising/falling projected outer profile; glass behind a dense timber screen. Views 01–04/07/08 show a side wave from straight slats with layered depth and shadow. Retain. The outer articulation is transferred; glass, dense screening, vegetation and optical performance are outside the adopted brief.

FRIM Sheet 9 truss elevation / heel. Slender triangular truss with heel supports and center rise. Views 08/11/12 show changing-depth roof triangles and lower-chord seats on paired column heads. Retain topology and supported heel. Faceting, span and paired nodes are project-specific rather than a scaled structural design.

FRIM Sheet 9 roof-bracing plan. Roof-plane diagonals restrain and tie the truss series. Views 08/17 show longitudinal ties and crossed diagonal pairs connecting the seven frames. Retain. Apparent crossings in projected images are not an identified solid penetration; the separate geometry report remains zero pairs. Brace joint capacity is unverified.

FRIM Sheet 9 gussets / nailing. Plywood gussets and specified nail patterns transfer node forces. Views 12/13 show layered timber contacts and straps without gussets or hardware. Retain as the explicit all-slat deviation. This does not establish that the substituted node can develop the source truss forces; fastening must be designed.

CMHC Fig. 68 wall opening. Lintel and jamb framing carry interrupted wall infill. Views 06/15/19 show the supported entrance header and jambs with infill above. Retain opening grammar. The Builder reports 90 mm seats; the render cannot independently measure all bearing areas or verify connections.

CMHC Fig. 86 rafter heel. Roof members seat over aligned supports. Views 11/12/16 show roof lower-chord seats at the side-column heads and the level ceiling below. Retain aligned bearing logic. The proposed truss does not adopt rafter/collar-tie prescriptions. Ceiling gravity still requires the named face fixings.

CMHC Fig. 47 sleeper on slab. Supported timber sleeper with separate ground/moisture layers. Views 14/18 show the crossed slat mat and continuous sole runners. Retain geometric bearing arrangement, with no claim that the omitted slab, moisture break or anchorage is supplied by wood-to-ground contact.

Disposition: all five primary photo rows and all six cited manual-figure rows are retained or intentionally synthesized with the reasons above. No concrete missing member, unintended envelope gap, obstructed opening, wrong profile or other geometric/photo defect was identified. R-13 is checked, bringing requirements to 28 of 28 without waivers. Phase 2 accepts v02 unchanged; another geometry version would have no review finding to resolve.

Incidental specification correction: the existing roof formula samples seven ribs at three repeated vertical offset levels through one rise and fall. The earlier phrase "seven distinct rib heights" overstated what that formula says. The phrase is corrected in concept.md; the formula, dimensions and built geometry are unchanged.

Review cost evidence: one gpt-5.6-luna comparison Inspector, five primary photos, six manual snippets and seventeen v02 render views. This Designer reviewed built notes and reports without opening scripts or renders. No additional renders, manual research, geometry files or shared-tool changes were made during this phase. Token and monetary totals are not exposed and are not estimated.

Experiment 16 version notes — GPT-6

v01 — phase 1, 2026-09-14

Implemented the released 3.000 × 3.000 × 6.000 m clear room with one 0.900 × 2.100 m entrance, two open high clerestories, flat red vertical boarding and a level timber ceiling. Seven transverse frames carry paired pale timber chords, red web triangles, varying outer-column profiles and a fully boarded faceted roof. The geometry starts from tools/experiment_template.py and uses craftbot lib, geometry2d, planes and framing; no shared toolkit was changed.

The crossed ground mat includes the Designer-approved continuous sole runners within its upper layer, z=−0.060..−0.024. Transverse sleepers butt to those runners, with lower longitudinal sleepers directly below their centerlines. A ripped edge slat carries both entrance-jamb plies. The doorway header has 90 mm seats. Side-wall rails are ripped to the specified 31 mm attachment gap; end rails terminate against the inner-chord faces. Wall butt joints stagger on rail heights and individual wall boards are at most 3.000 m long.

Column-chord turns use complementary mitres. Roof chords are split at web nodes; their upper faces bevel to the additive roof facets. Short B roof boards span adjacent 0.500 m frame stations and meet on complementary vertical facet boundaries. Ceiling boards are face-fixed to the lower chords; concealed fixings carry their gravity load; the boards do not bear on the tops of members above them.

Longitudinal ties and end-bay diagonals connect the transverse frames. Roof-plan crossed braces occupy separate 36 mm layers between the lower-chord faces. Ground-plan diagonals fit between the crossed sleeper network in its two thickness layers. These members name the lateral load path; boarding alone is not called a diaphragm, and the concealed connections remain undesigned.

Verification

– Final rendered model: **2,101 members, 0 penetrating pairs >1 mm, 0 floating members at 2 mm contact tolerance, 0 overlap families.**

– No ignored contact families and no changed check tolerances.

– Initial unrendered preflight: 2,031 members, 86 pairs, 0 floating. Its two causes were longitudinal ties crossing column runs (70 pairs) and longitudinal diagonals crossing transverse column webs (16 pairs). Fitting these members to the web faces removed both causes; the segmented ties add 70 members. Both preflights occurred before the first render, so there is only one rendered v01.

– Script assertions check clear room and entrance volumes against every member's bounds, exact room ratio, frame spacing/count, clerestory limit, roof-facet count, declared catalogue/material, floor support spacing, sole/support centerline alignment, header seats and rail spacing.

– The full renderer ran successfully in Blender 4.3.2; all views **01–19** were rendered. Views include four orbits, top/front/east elevations, frame-only, from below, room section, roof/profile, heel, layered column node, base, entrance, roof boarding, roof-plan bracing, mat

underside and interior front.

– Saved model: experiment_16_gpt6_v01_blender.blend; pair report: experiment_16_gpt6_v01_blender_pairs.txt. The version is immutable after these renders.

– Closeout's textual view-schema check initially missed the valid frame-only view because it recognizes only a double-quoted inline hide list or a variable named COVER. View 08's identical hide values were inlined using double quotes; camera, visibility and rendered view identity did not change. No render was necessary.

Bearing audit beyond the contact check

R-16 remains open. A horizontal cut through each sloping outer-column foot produces a horizontal width of **90.008–91.505 mm**, against the 90.000 mm sole/support width. The maximum unsupported edge is **0.753 mm on each side**. The contact check correctly reports contact but cannot prove full footprint bearing. The next version must add adjacent ripped S sole/lower-support strips with fitted transverse-mat cuts, or fit a local foot-end bevel within the S stock. Preserve the released column profile, mat datums and clear-room dimensions.

Inspection and requirements

The fresh single-room Inspector reviewed all 19 views and reported no clear major missing geometry, unintended envelope void, misplaced member or floating member. The room, entrance, upper light, seven frames, layered chords and expressive roof are visually confirmed. Detailed observations are in inspection_v01_room.md; the merged result and Builder dispositions are in inspection_v01.md.

R-14, R-15, R-21, R-22 and R-23 contain details that cannot be measured or seen fully in these renders. Their generation rules, dimension chains, fitted-plane construction and named ceiling face fixing supply the Builder evidence; there is no contrary visual finding. The requirement checkboxes now record **26 of 28 satisfied**: R-13 awaits phase-2 reference comparison, and R-16 remains open. Thus **phase 1 is not converged**. No other visual correction was requested for v02.

Scope and evidence limits

CMHC board-thickness examples and the FRIM slender-truss drawing are precedents, not capacity evidence for this assembly. The 6 m slat columns are outside the cited prescriptive stud cases; FRIM plywood gussets do not validate the modeled paired-slat joints. No plywood, metal shoes, glazing, concrete or invented heavy beams are modeled. The low threshold replaces the reference stair, the usable doorway replaces its unusually low aperture, and the exact 2:1 clear height follows the user's brief.

This is a pavilion with open high clerestories. Structural adequacy, fastener and connection design, uplift/overturning anchorage, weatherproofing, durability, fire, soil bearing and moisture separation are not verified. Sections are plausible visual propositions, not calculated. These limitations do not waive geometric bearing or inspection findings.

Next version and Designer questions

1. Correct the quantified R-16 foot-seat deficit with an in-catalogue local assembly detail; the two alternatives above preserve the design. No scope change is requested.
2. No additional visual defects were identified. Recheck the base close-up, full numeric model and required full phase-ending views in a fresh v02; do not overwrite v01.
No open design-authority question at this stage. Usage/context statistics are not exposed to this Builder and are not estimated.

v02 — phase 1 foot-bearing correction, 2026-09-14

Copied the immutable v01 script and corrected R-16 with fitted foot-end bevels. Each sloping first chord segment has two additional cut planes: they run from the edges of the 90 mm horizontal seat to the original stock edges 90 mm above that seat. Above this short end detail the original chord profile remains. The cuts remove material from the S36x90 blank, keep every solid convex, and retain the fixed column centerlines, room dimensions, sole runners and ground-mat datums. No support strips or other members were added. The alternative of adjacent ripped supports would require new mat cuts and more pieces; the fitted bevel resolves the local deficit directly.

Verification

– **2,101 members; 0 penetrating pairs >1 mm; 0 floating members at 2 mm contact tolerance; 0 overlap families.** No ignored members or changed harness tolerances.

– A new assertion selects the four actual mesh vertices of each horizontal chord foot and checks that its full rectangle lies within both the sole-runner and lower-sleeper footprint. It also checks the support top levels and requires a full 90 mm seat. The 1 micrometre assertion tolerance accommodates Blender mesh precision; it is distinct from the overlap/contact tolerances.

– Test-first evidence: before the bevel was added, that assertion failed on ColumnOuter -1 00 -1 00 against SoleRunner_00. After the correction, all **56 foot faces** passed, with measured widths **90.000–90.000 mm**. The previous maximum 0.753 mm unsupported edge is removed.

– Full Blender 4.3.2 render completed with exit code 0; every stable view **01–19** was rendered. No view definition changed. The base is shown by views 14 and 18, with frame and underside context in 08 and 09.

– Artifacts: experiment_16_gpt6_v02.py, experiment_16_gpt6_v02_blender.blend, experiment_16_gpt6_v02_blender_pairs.txt, and 19 experiment_16_gpt6_v02_blender_view_XX.png renders. No changes to v01 geometry or shared tools were made by this Builder. Runner owns viewer export and close-out.

Inspection and requirements

The fresh single-room Inspector completed all 19 views and found no missing member, unintended envelope gap, clearly floating member or new absence. Base view 14 shows seated feet without a visible gap; the small bevel itself is below reliable visual measurement, so the full-foot assertion supplies the exact support evidence. Frame, underside, heel, node, roof-board and bracing views show no regressions. Its saved report inspection_v02_room.md was complete before the final-message usage interruption; the continuation on 2026-09-15 merged it into inspection_v02.md without repeating renders or inspection.

R-16 is now checked. **27 of 28 requirements are satisfied, and phase 1 is converged**: 0 pairs, 0 floating, no open visual defect, and every phase-one requirement satisfied. R-13 alone remains pending for phase-2 reference comparison. Runner's closeout_v02.md records **8 passed, 0 failed**, including exported model, layers, all 19 renders and viewer screenshots.

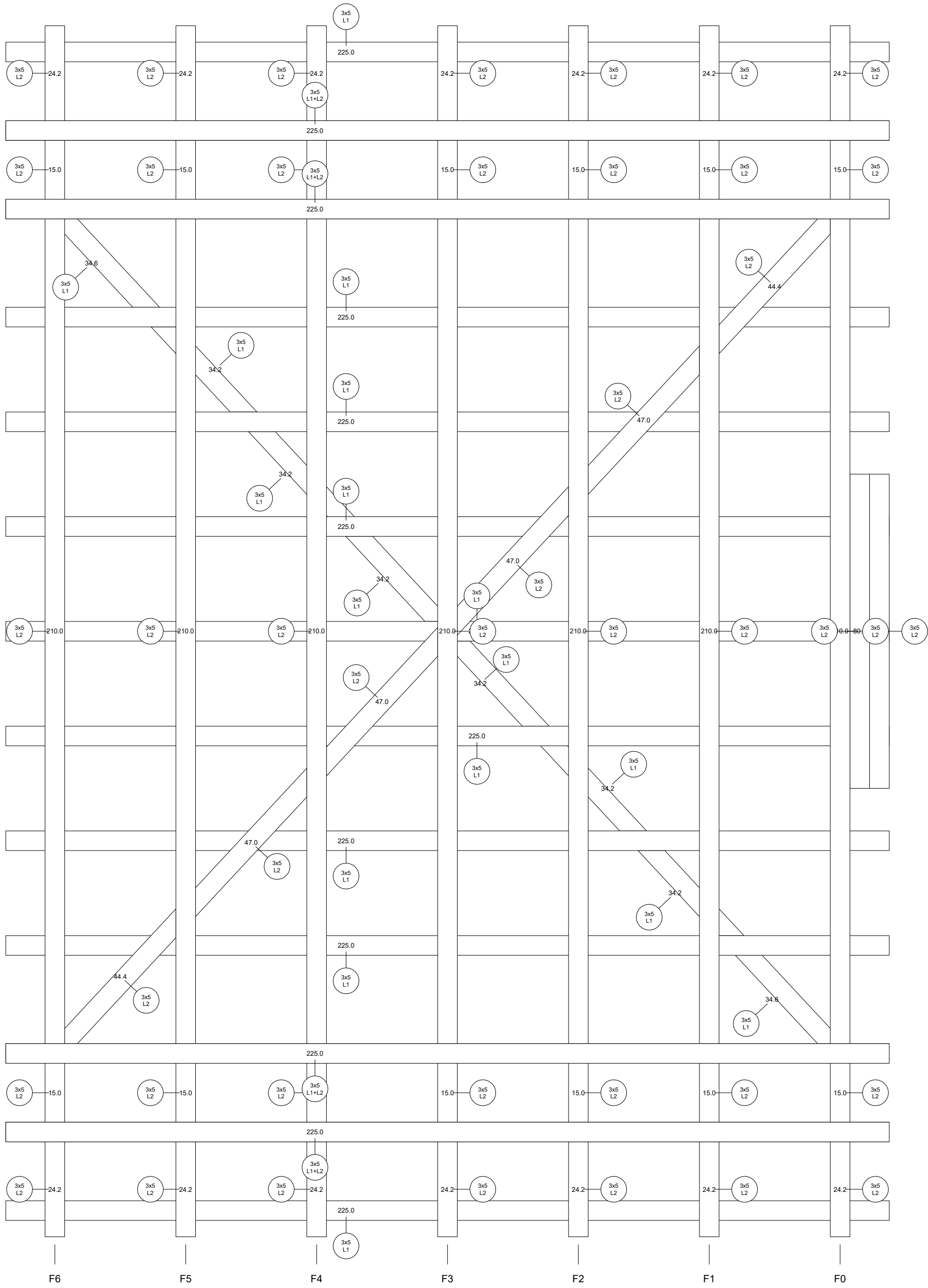
Scope and Designer questions

The named load path remains chord foot → continuous upper sole → directly aligned lower sleeper → grade. Full geometric bearing is verified; fastener forces, buckling, capacity, anchorage, soil/moisture design and weatherproofing remain unverified. The open clerestory pavilion and all v01 scope interpretations remain in force. Sections are a visual proposition, not calculated.

No open design-authority question. No shared helper promotion is proposed. Usage/context statistics are not exposed to this Builder and are not estimated.

Credits

The project author is Luka Piškorec, previously a lecturer at Aalto University and researcher at ETH Zürich, co-founder of TEN Studio (Zürich and Belgrade) and {protocell:labs} (Helsinki), practices that work at an intersection of architecture, design, digital art and research. CraftBot is part of the art-ai-fact initiative funded by Aalto University in 2025-2026.



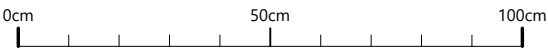
NOTE Two crossed sleeper layers with sole runners and plan braces. Door end at the bottom. Numbers: cut length in mm. Circles: stick profile and layer, L1 lies on the paper.



DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

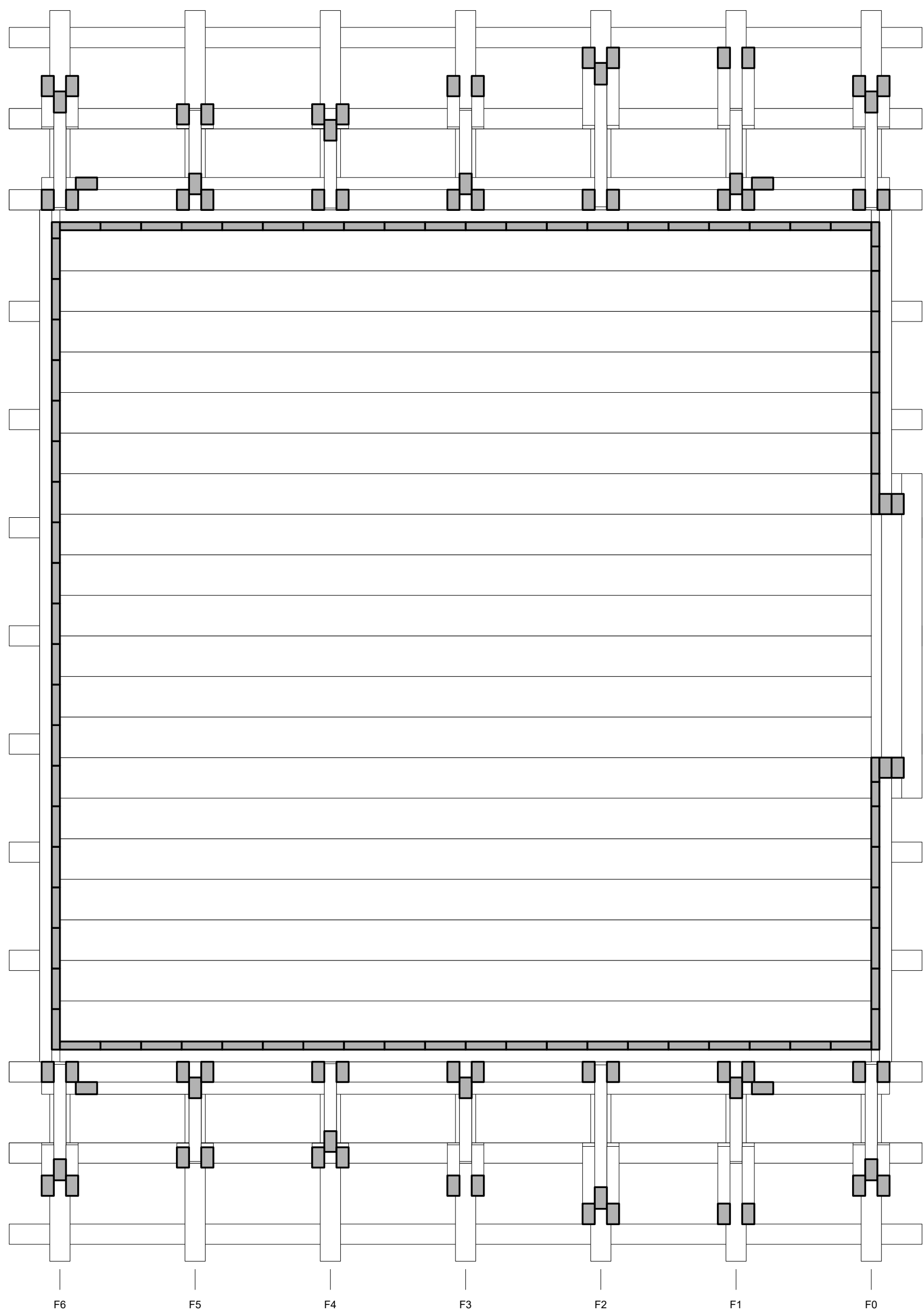
DISCLAIMER This is an AI generated project. 3D model is reproducible from Python code. Drawings are derived from a 3D model and exact. Minor geometric and constructive inconsistencies possible on the 3D model level.
SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE Plan, ground mat
PROJECT Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET 03



NOTE Horizontal section 80 mm above the floor, looking down. Cut members shaded.



DATE	2026-09-25
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AGENTS	CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
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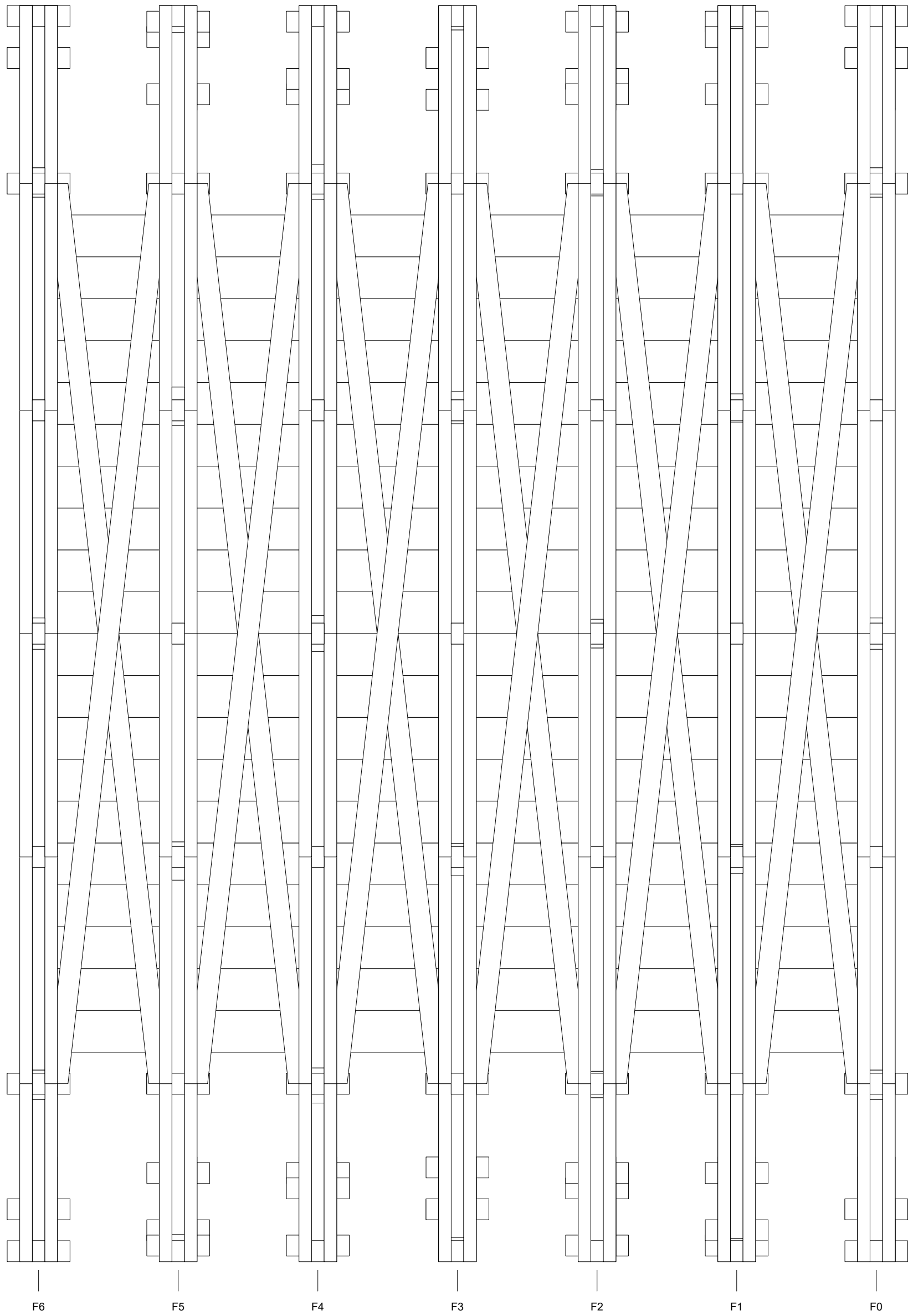
SCALE
1:15, drawn 1:1 to the model, units mm



TITLE
Plan, section at +80 mm
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
{protocol:labs}

SHEET

04



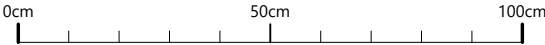
NOTE Roof boards removed. Crossed plan braces lie between the lower chords.



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AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

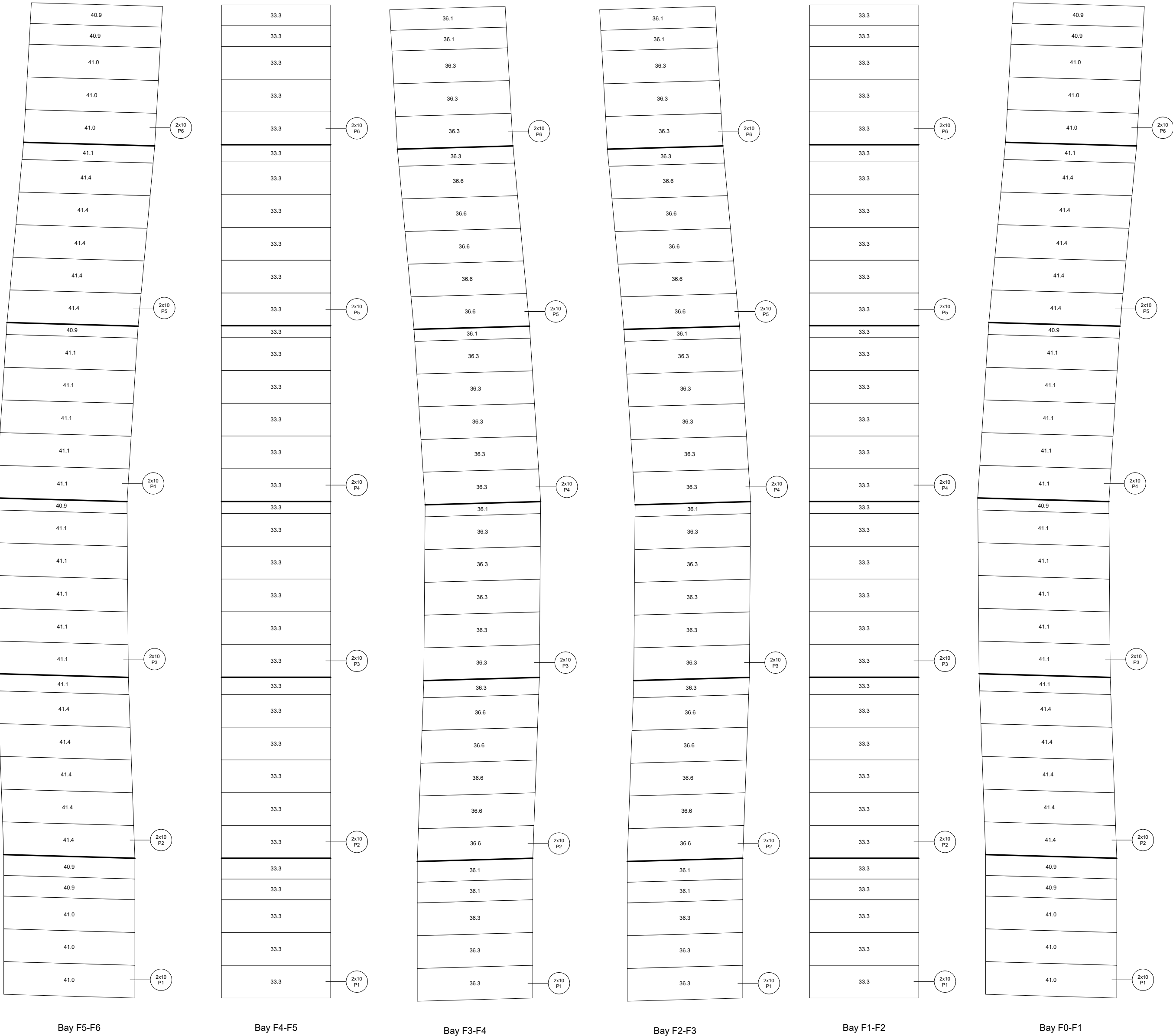
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Plan, roof framing
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol : labs

SHEET
05



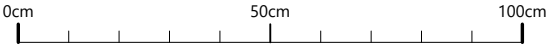
NOTE Roof boards of each bay laid flat, door end at the bottom, -x on the left. Heavy lines: folds between the six planar panels P1 to P6 of a strip. Numbers: cut length in mm. All boards 2x10, one layer.



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WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

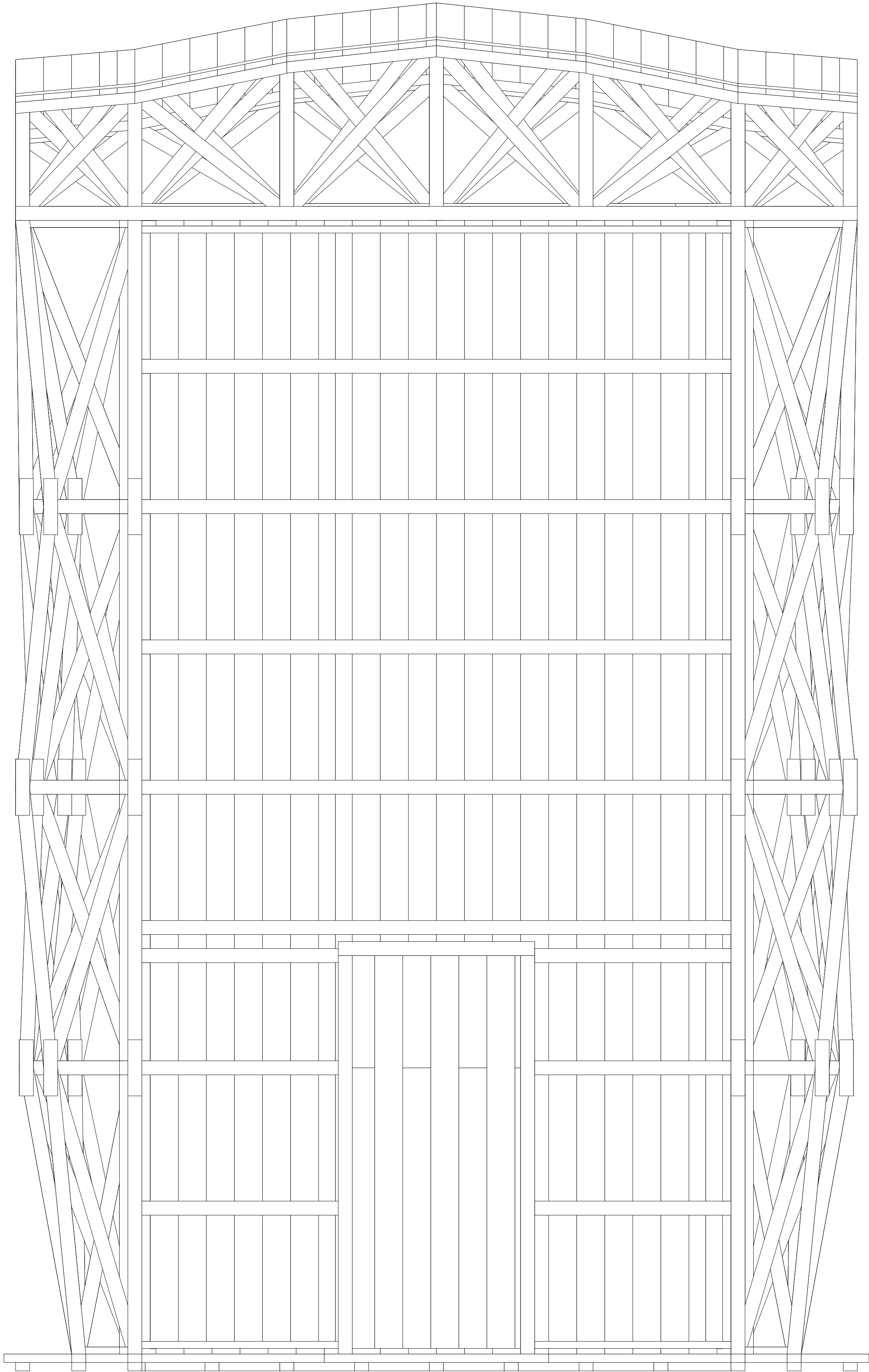
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Roof boards, unrolled
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET
06



DATE 2026-09-25

SUPERVISION Luka Piškorec, MSc ETH Arch

AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner

WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE
1:15, drawn 1:1 to the model, units mm



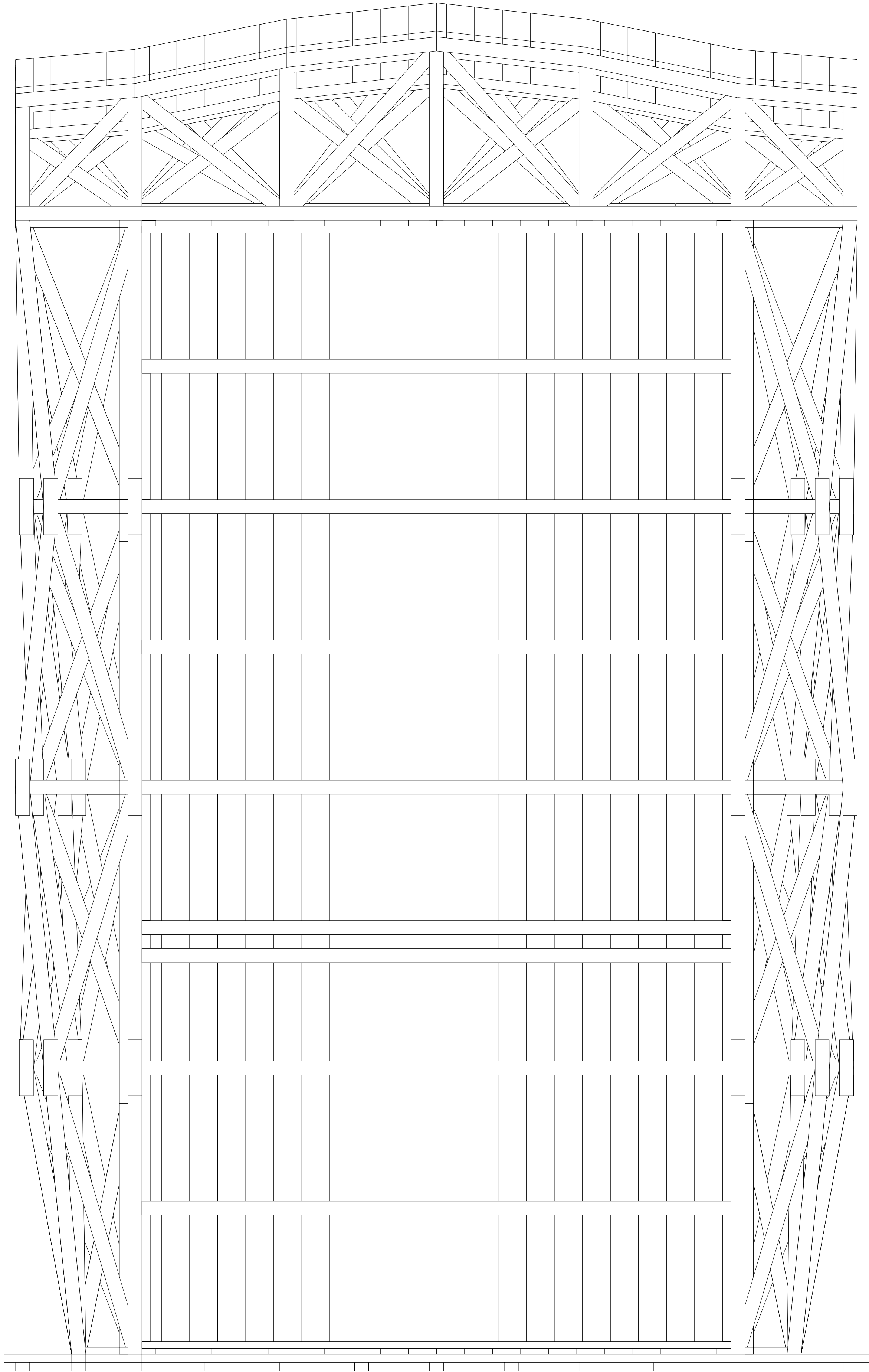
TITLE
Elevation, door end

PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02

protocol:labs

SHEET

07



DATE 2026-09-25

SUPERVISION Luka Piškorec, MSc ETH Arch

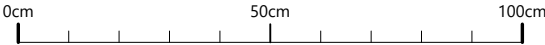
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner

WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm

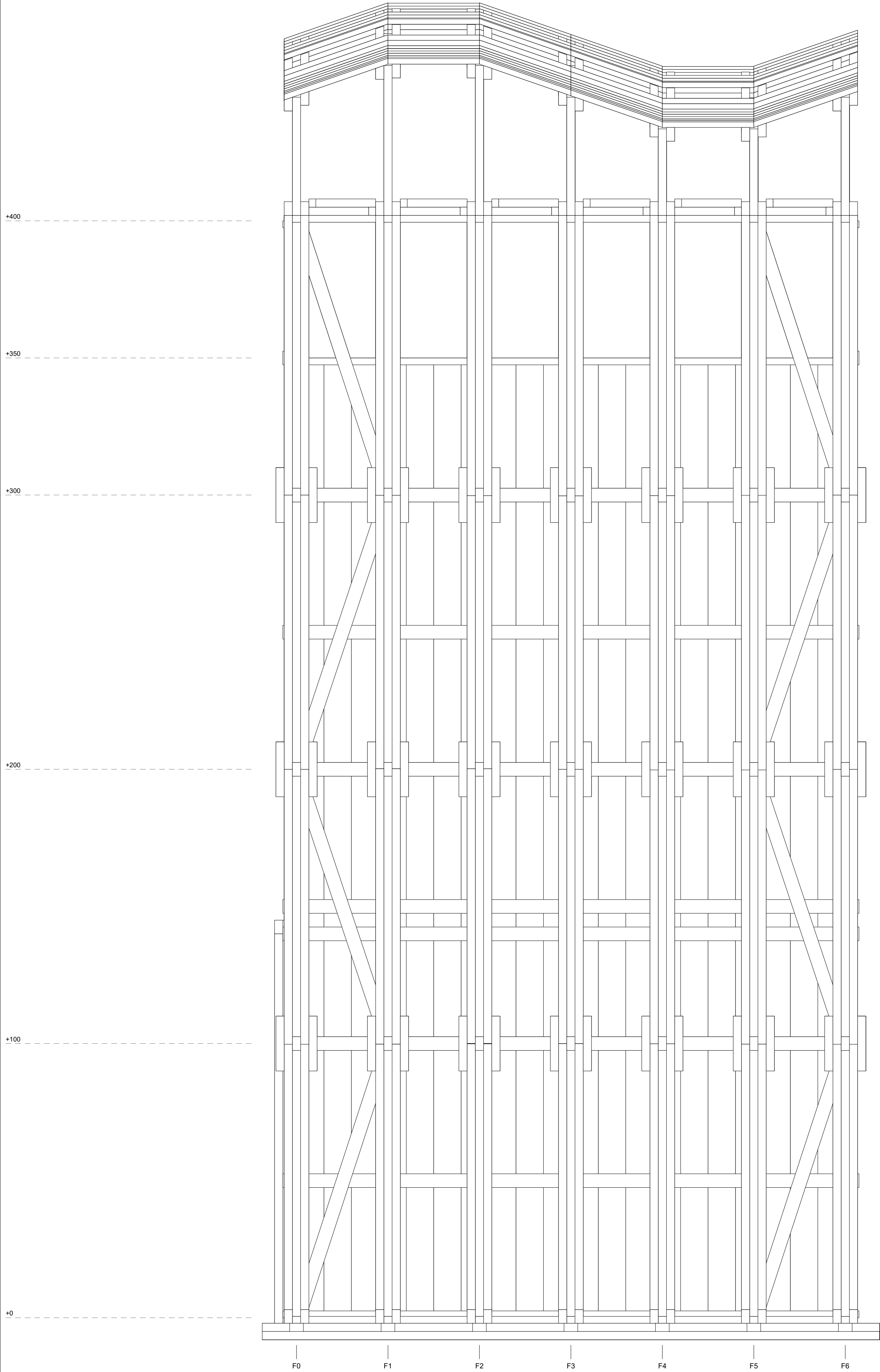


TITLE Elevation, far end

PROJECT Experiment 16 - Expressive Structure, GPT-6 v02

protocol:labs

SHEET 08



DATE 2026-09-25

SUPERVISION Luka Piškorec, MSc ETH Arch

AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner

WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

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SCALE 1:15, drawn 1:1 to the model, units mm

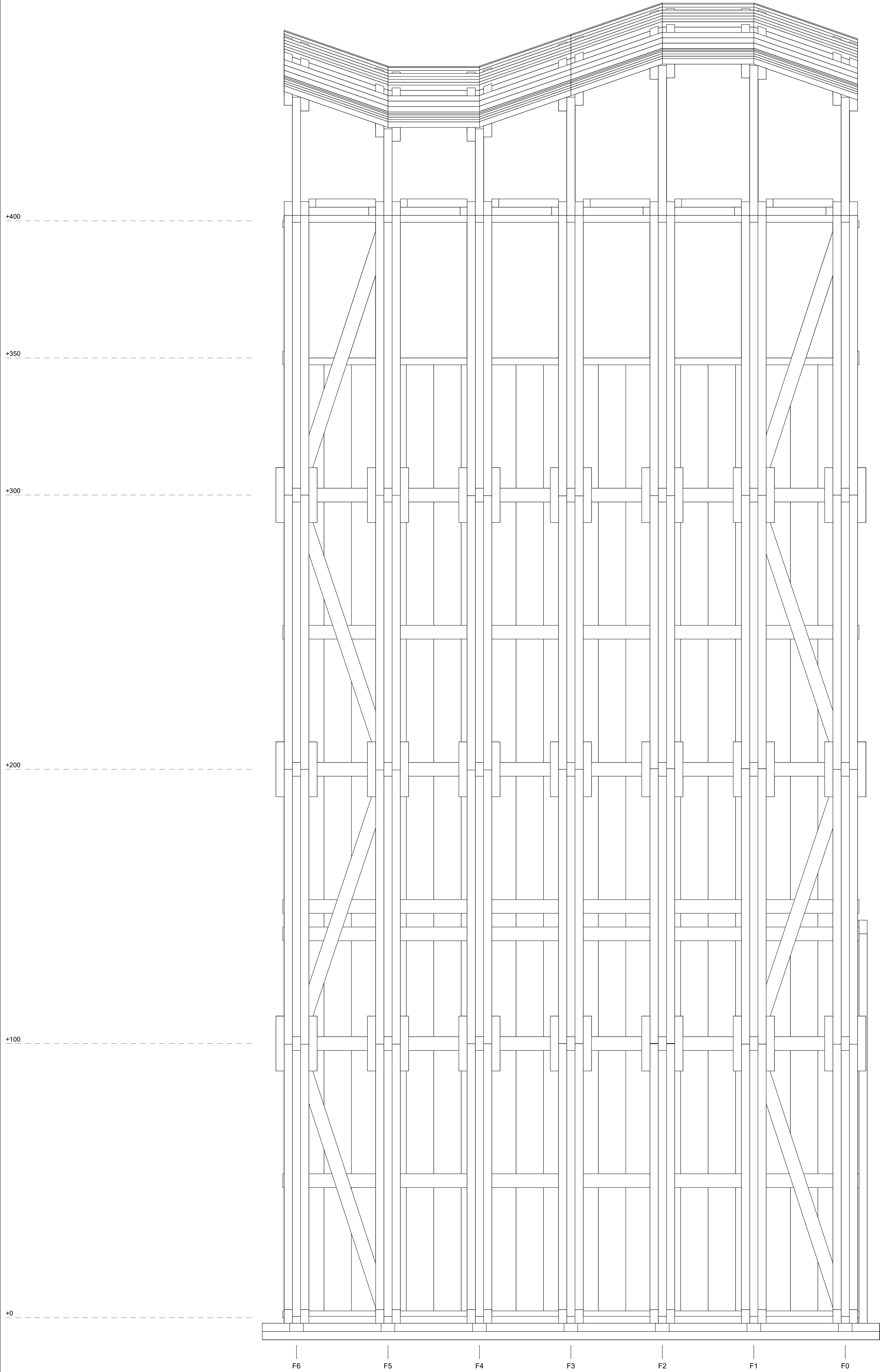


Elevation, side plus x

PROJECT Experiment 16 - Expressive Structure, GPT-6 v02

protocol : labs

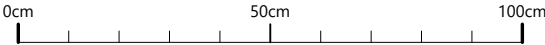
SHEET 09



DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

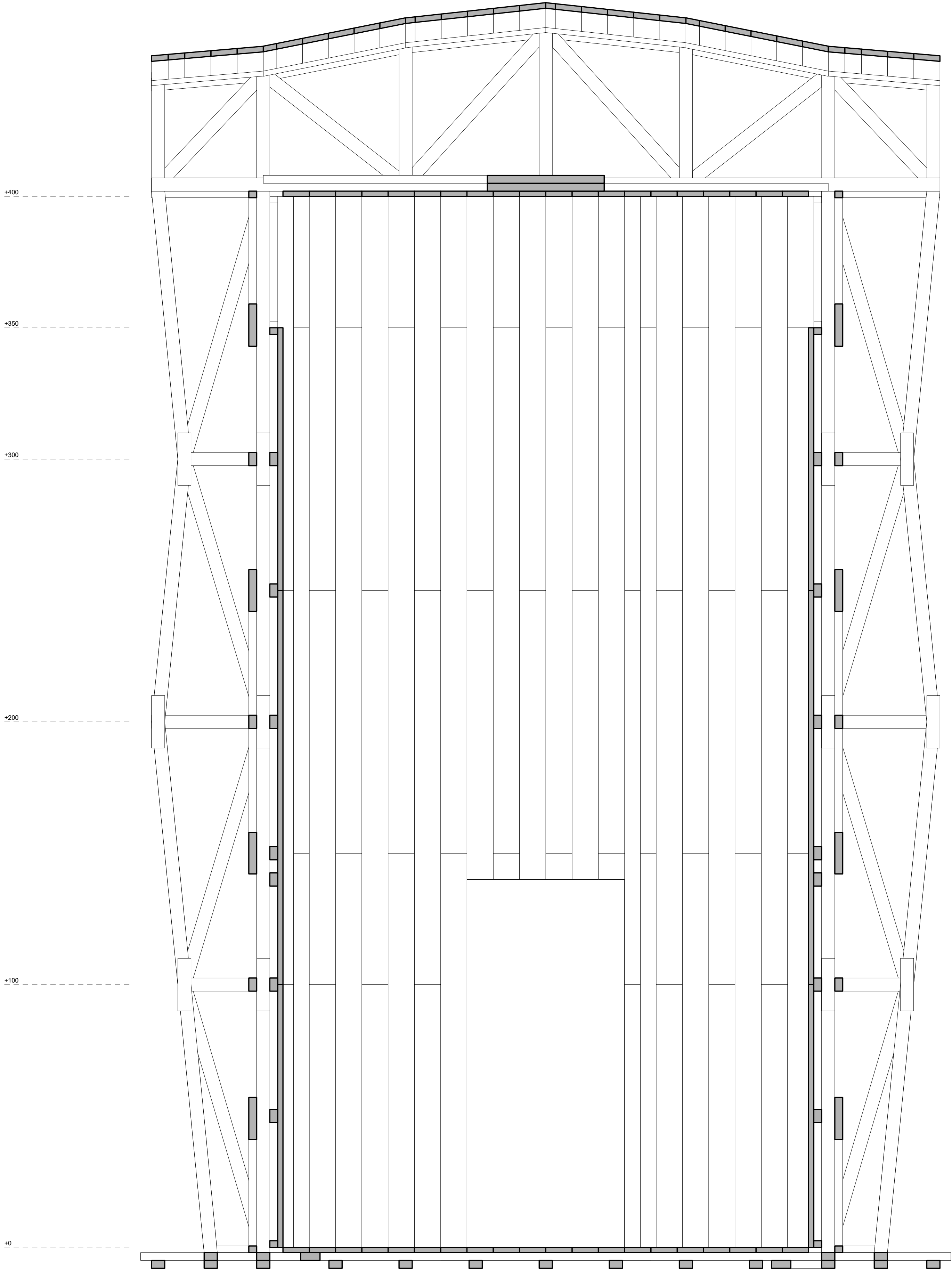
DISCLAIMER This is an AI generated project. 3D model is reproducible from Python code. Drawings are derived from a 3D model and exact. Minor geometric and constructive inconsistencies possible on the 3D model level.
SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE Elevation, side minus x
PROJECT Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET 10



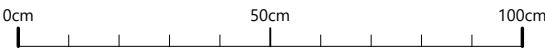
NOTE Cut midway between F0 and F1, looking towards the door wall from inside, so -x is on the right. Cut members shaded.



DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

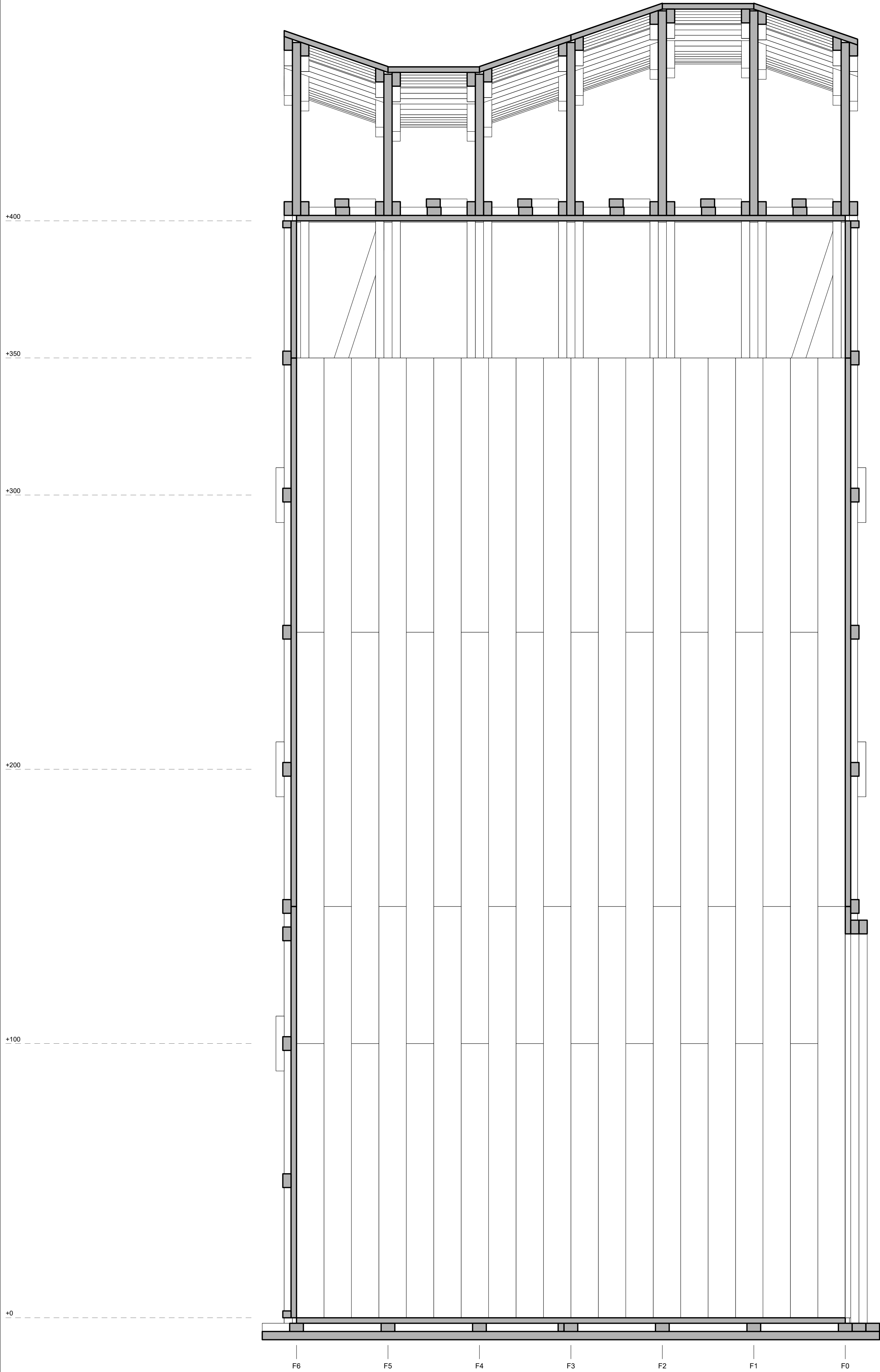
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



Cross section, first bay
Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET 11



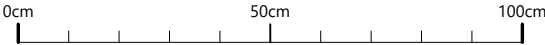
NOTE Cut 1.3 mm off the centre line so it passes through boards, not between them. Looking towards -x. Cut members shaded.



DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

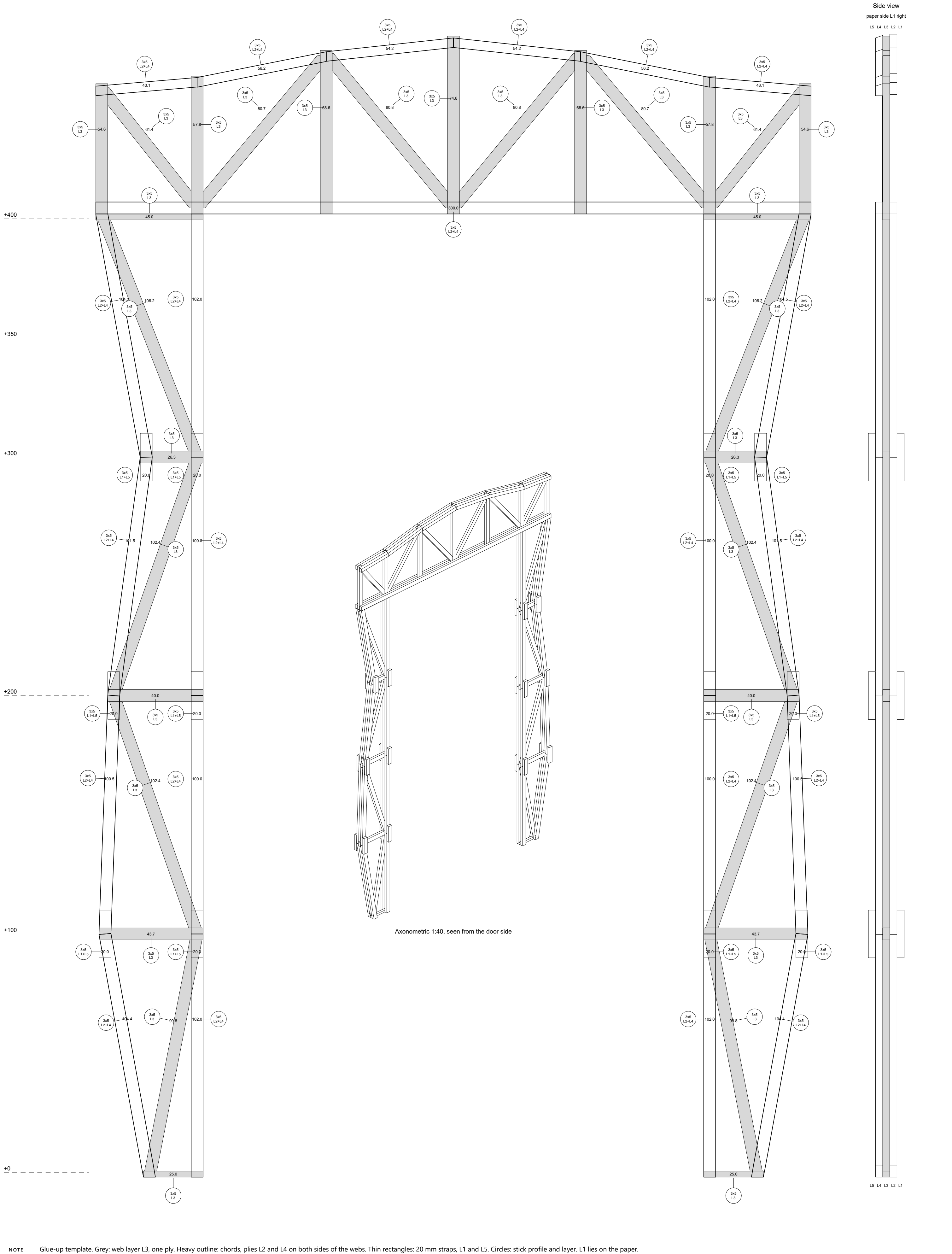
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE Long section, centre line
PROJECT Experiment 16 - Expressive Structure, GPT-6 v02
protocol : labs

SHEET 12



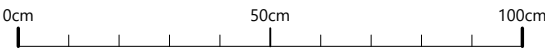
NOTE Glue-up template. Grey: web layer L3, one ply. Heavy outline: chords, plies L2 and L4 on both sides of the webs. Thin rectangles: 20 mm straps, L1 and L5. Circles: stick profile and layer. L1 lies on the paper.



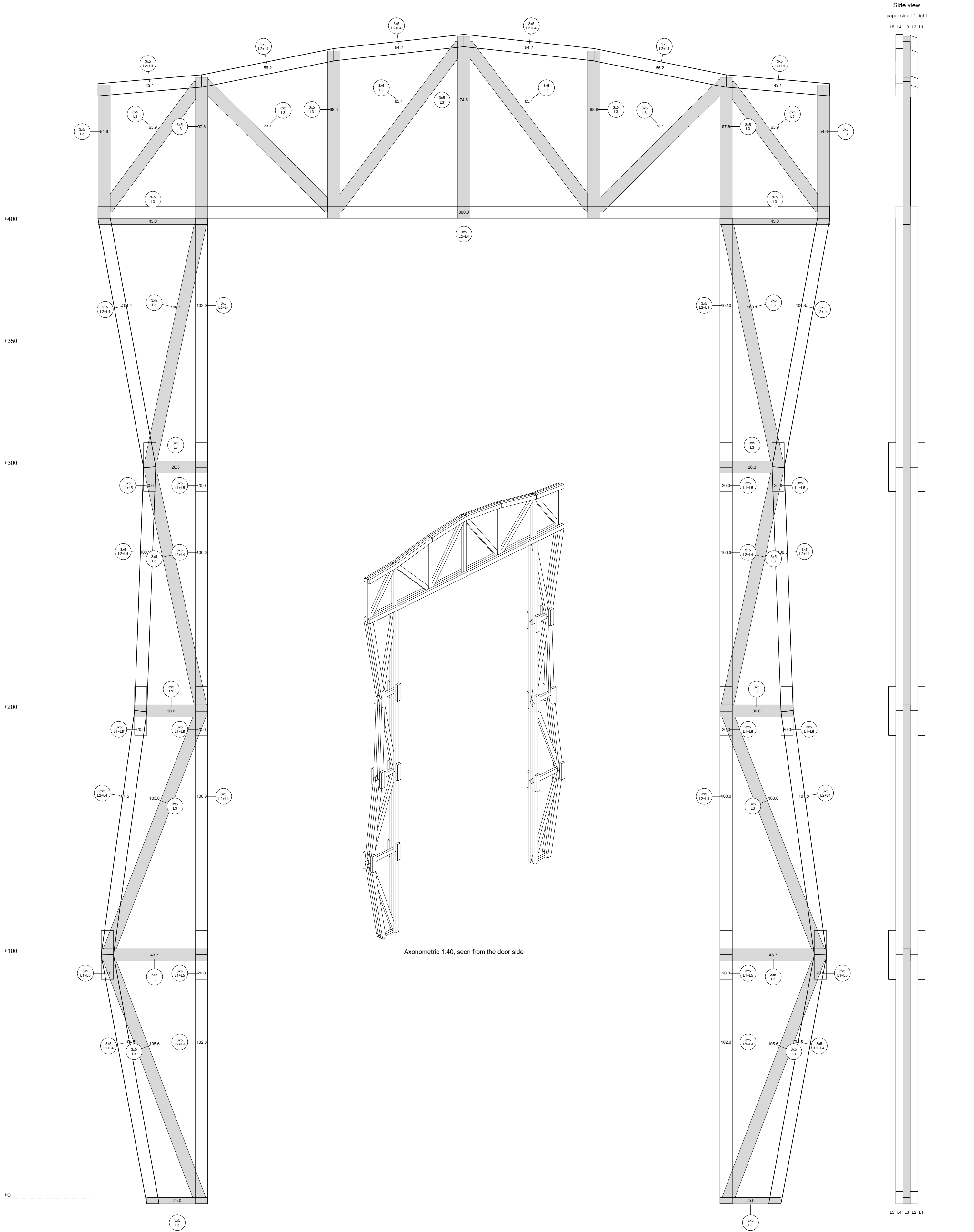
DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

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SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Frame F1 template
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
{protocol:labs}



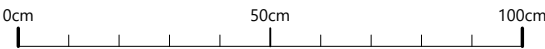
NOTE Glue-up template. Grey: web layer L3, one ply. Heavy outline: chords, plies L2 and L4 on both sides of the webs. Thin rectangles: 20 mm straps, L1 and L5. Circles: stick profile and layer. L1 lies on the paper.



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SUPERVISION Luka Piškorec, MSc ETH Arch
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WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

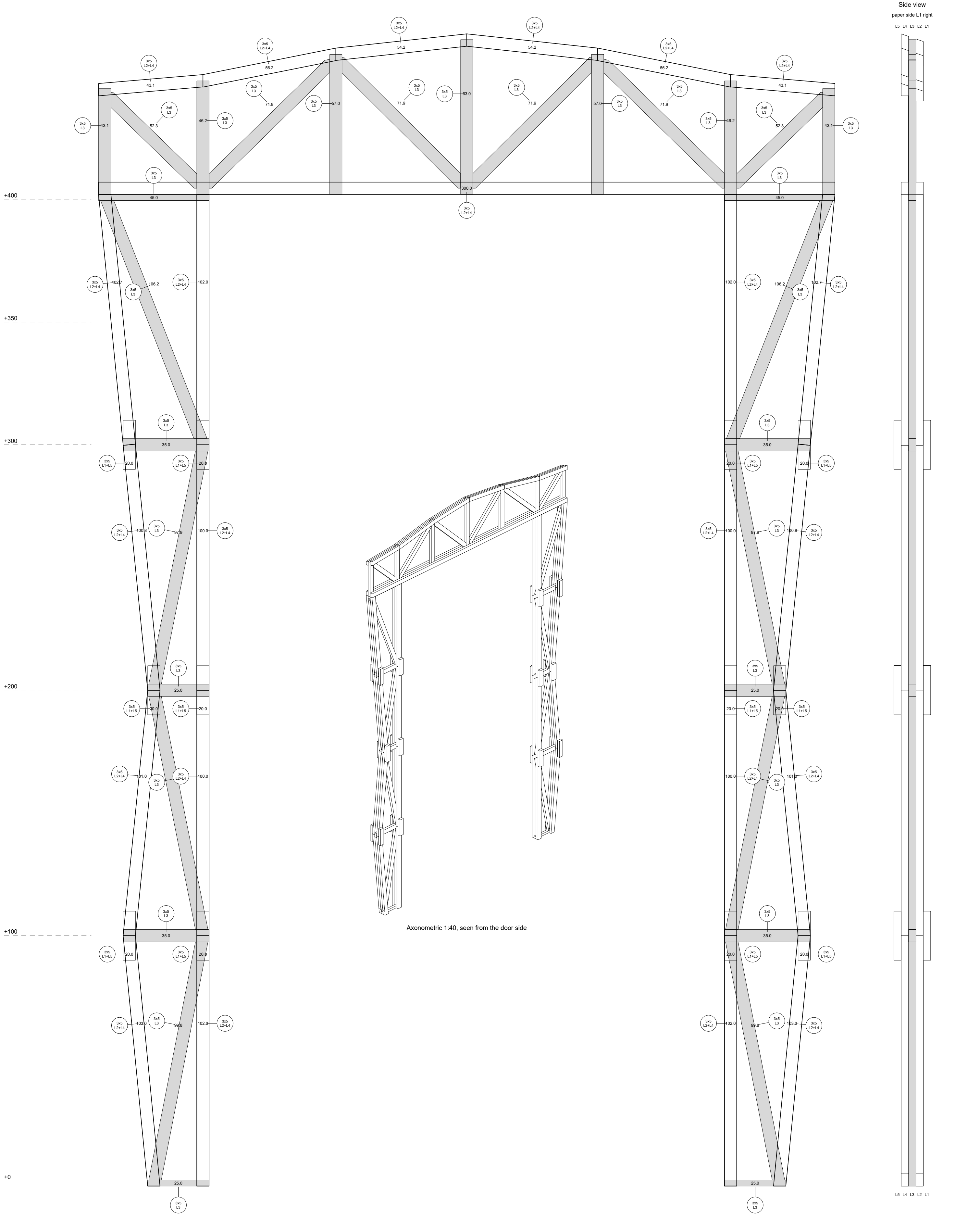
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

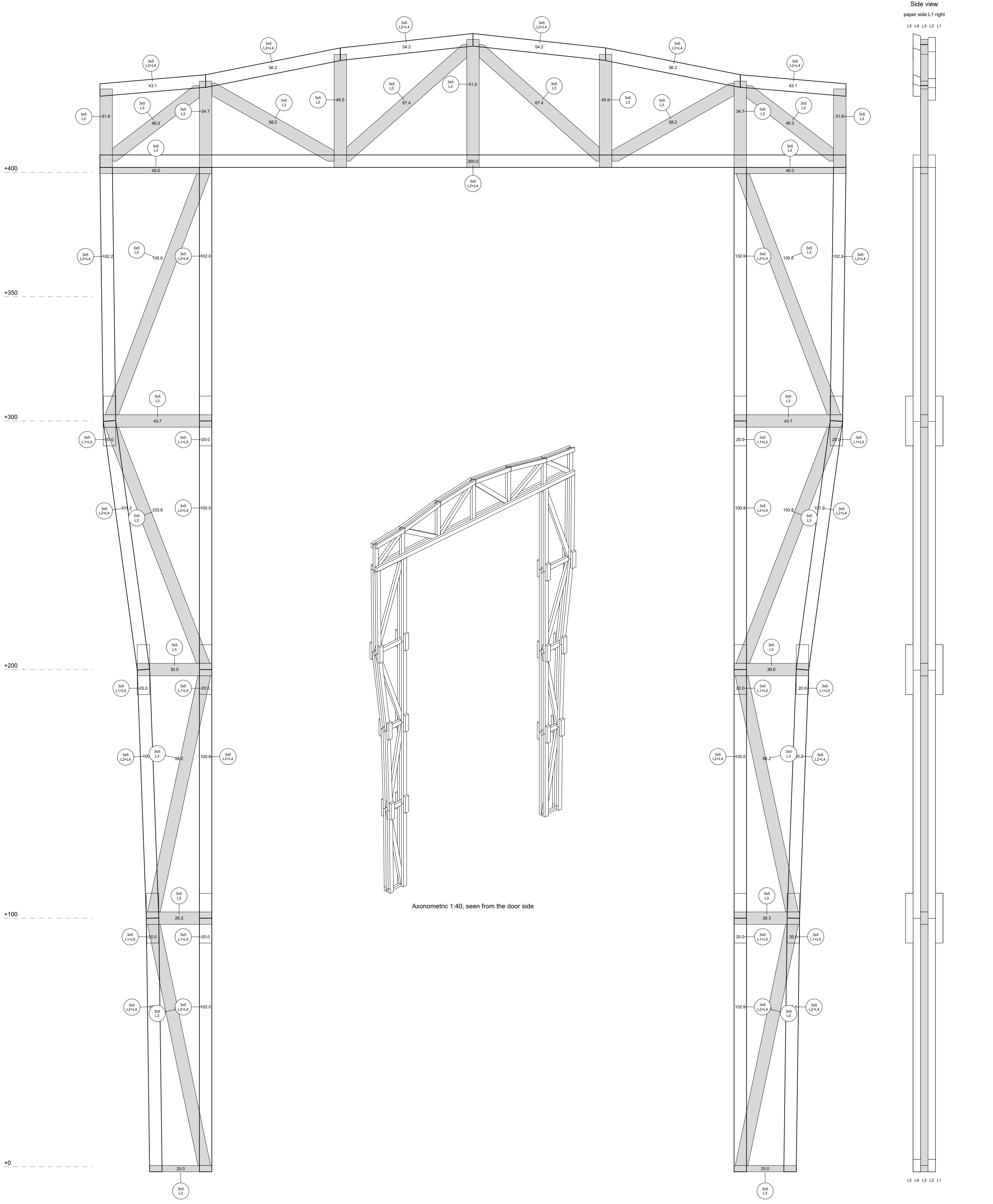
SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Frame F2 template
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
{protocol:labs}

SHEET
15





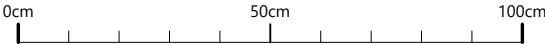
NOTE Glue-up template. Grey: web layer L3, one ply. Heavy outline: chords, plies L2 and L4 on both sides of the webs. Thin rectangles: 20 mm straps, L1 and L5. Circles: stick profile and layer. L1 lies on the paper.



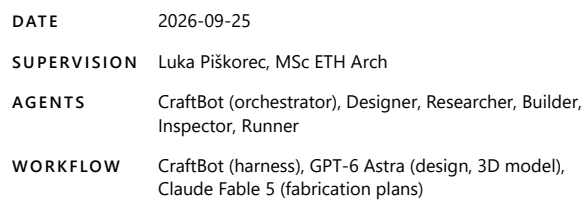
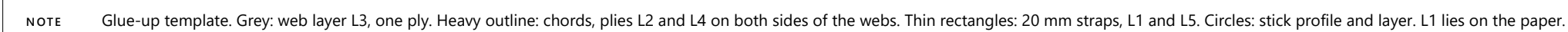
DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Frame F4 template
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
{protocol:labs}



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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE
1:15, drawn 1:1 to the model, units mm



PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02

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{protocell:labs}
```

EET

18

+400

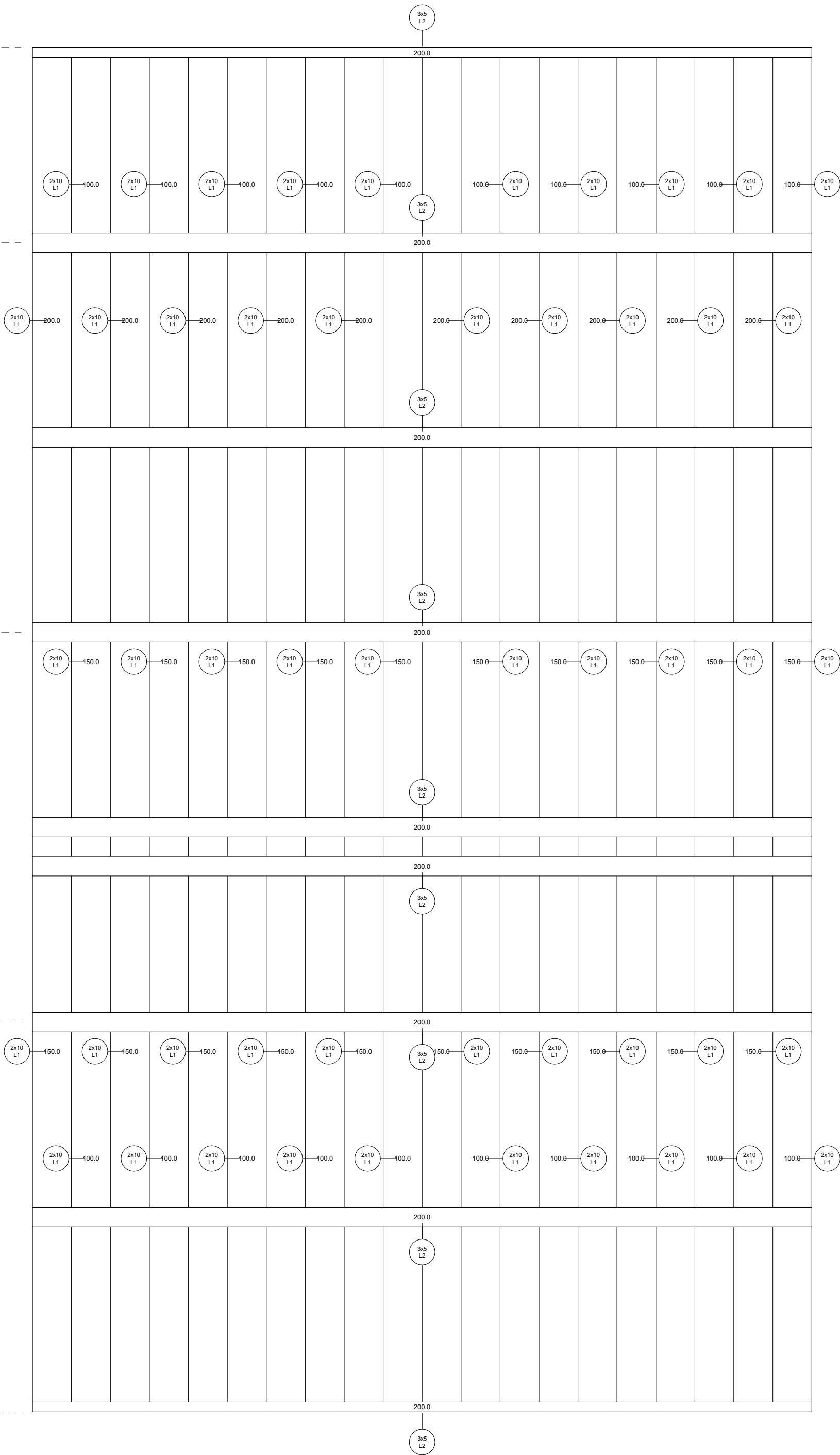
+350

+300

+200

+100

+0



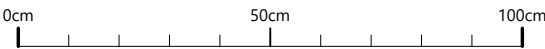
NOTE Glue-up template seen from outside: boards behind, rails in front. Numbers: cut length in mm. Circles: stick profile and layer, L1 lies on the paper. Build two, the second mirrored.



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WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

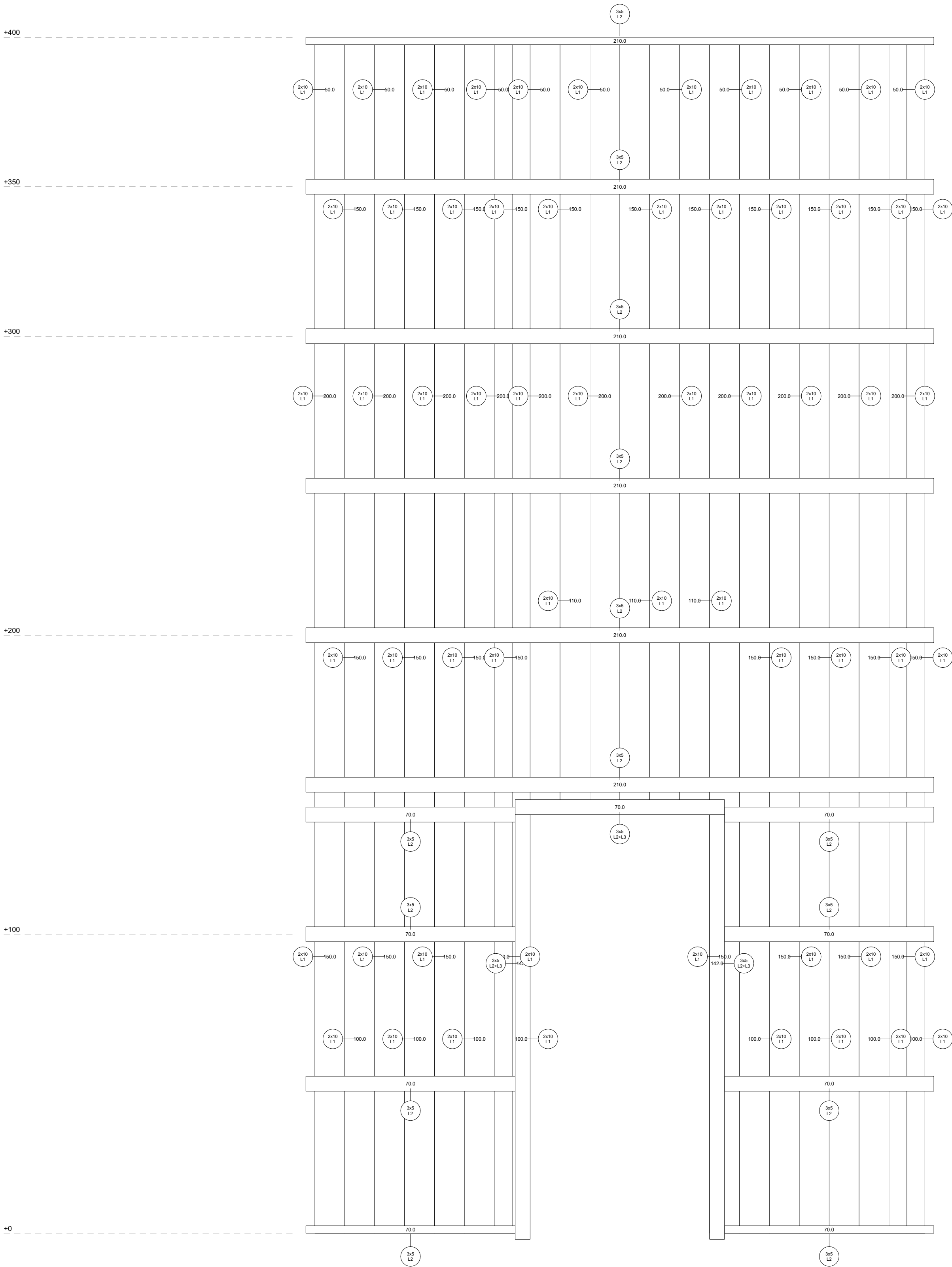
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE 1:15, drawn 1:1 to the model, units mm



TITLE Side wall template (both sides)
PROJECT Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET 20



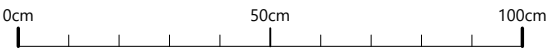
NOTE Glue-up template seen from outside: boards behind, rails in front. Numbers: cut length in mm. Circles: stick profile and layer, L1 lies on the paper.



DATE 2026-09-25
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WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

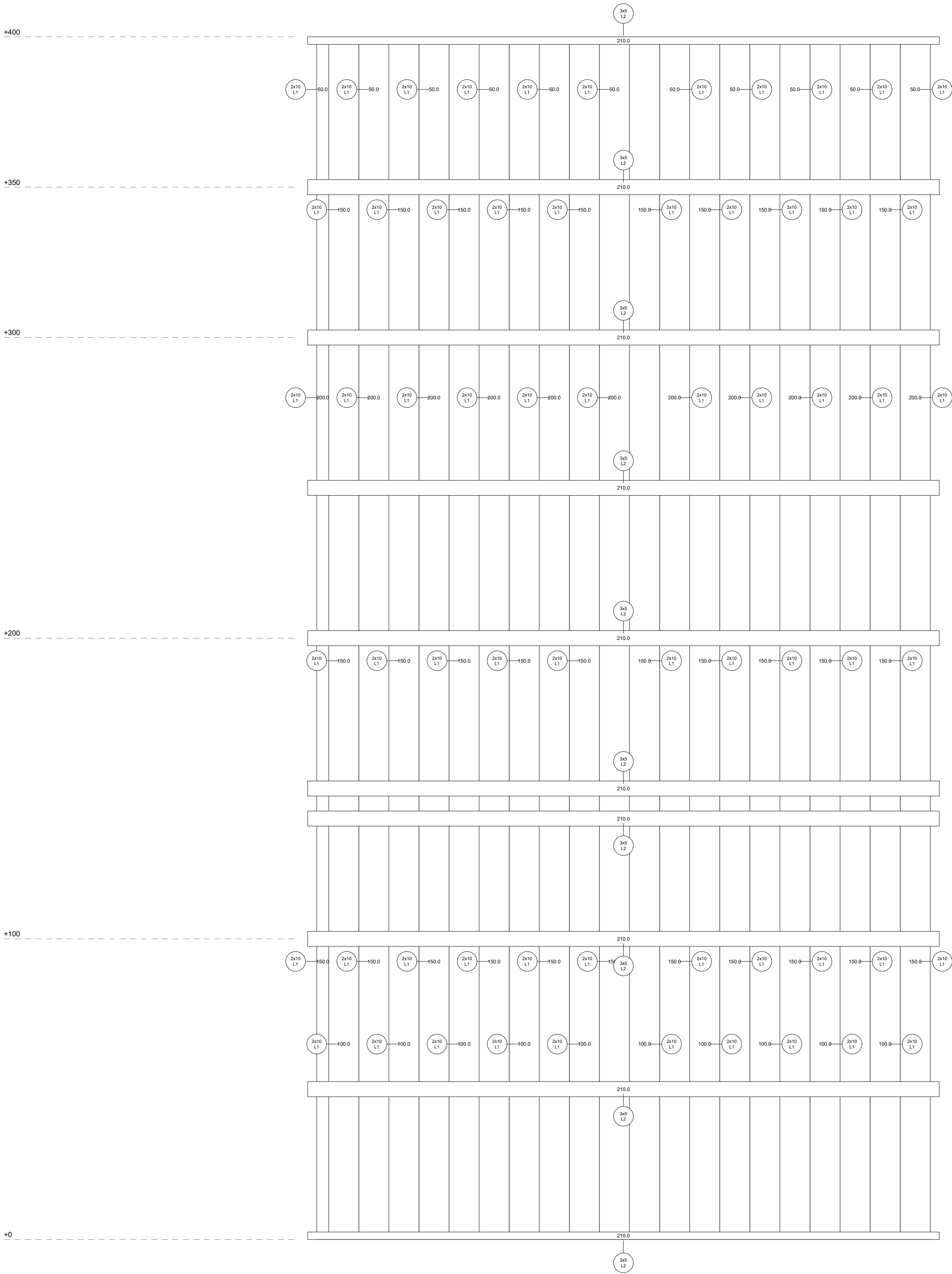
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SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Door wall template
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET
21

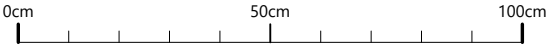


NOTE Glue-up template seen from outside: boards behind, rails in front. Numbers: cut length in mm. Circles: stick profile and layer, L1 lies on the paper.



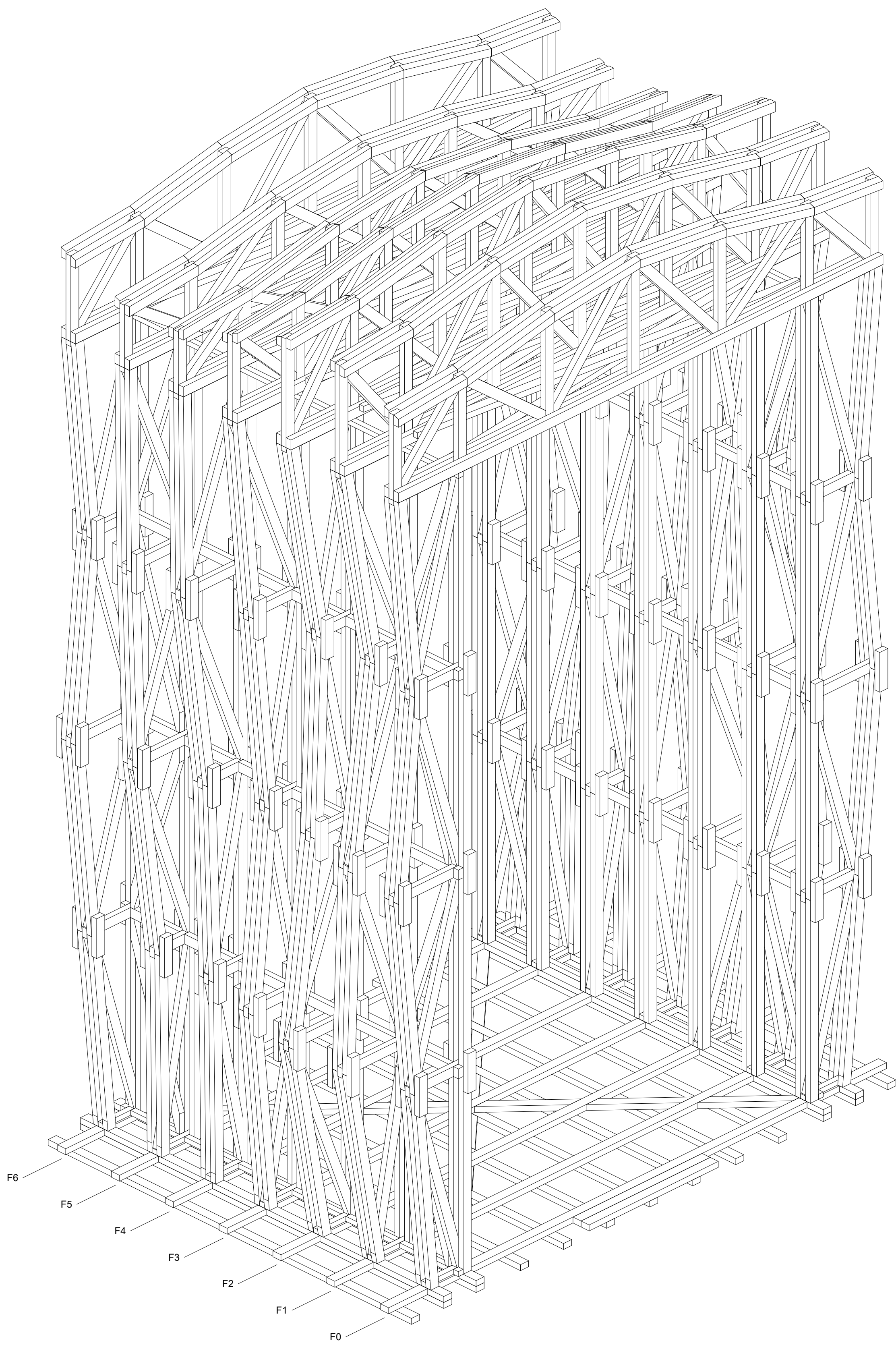
DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

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SOURCE CODE <https://github.com/lukapiskorec/craftbot>
SCALE 1:15, drawn 1:1 to the model, units mm



TITLE
Far wall template
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET
22



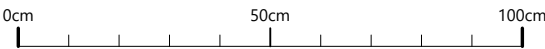
NOTE Every member of the viewer layer "frame". Seen from the door side.



DATE 2026-09-25
SUPERVISION Luka Piškorec, MSc ETH Arch
AGENTS CraftBot (orchestrator), Designer, Researcher, Builder, Inspector, Runner
WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

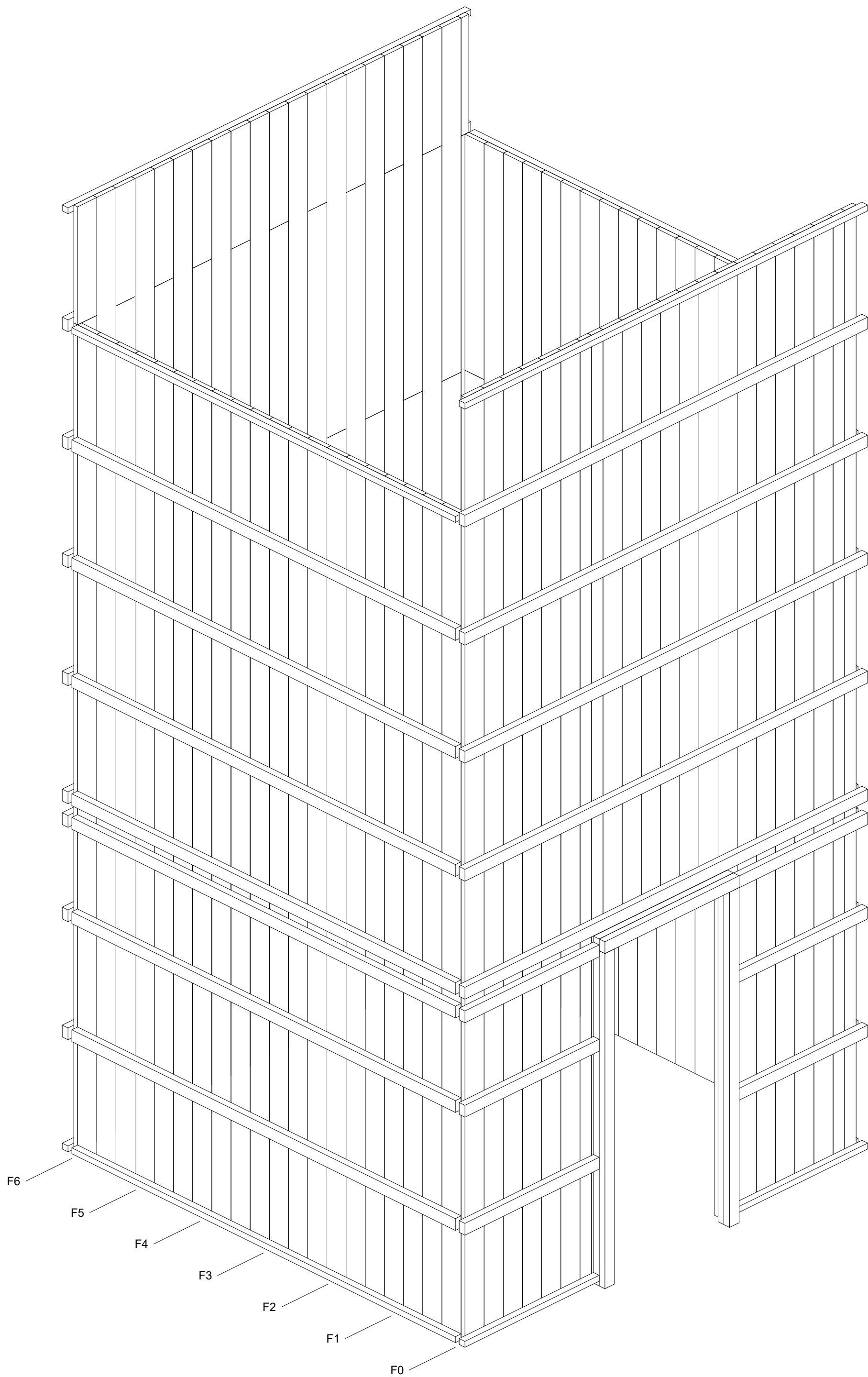
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SOURCE CODE <https://github.com/lukapiskorec/craftbot>

SCALE
Axonometric, scale 1:20, units mm



TITLE
Axonometric, frame layer
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET
23



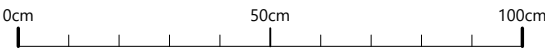
NOTE Every member of the viewer layers "cladding ext" and "fixtures". Seen from the door side.



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WORKFLOW CraftBot (harness), GPT-6 Astra (design, 3D model), Claude Fable 5 (fabrication plans)

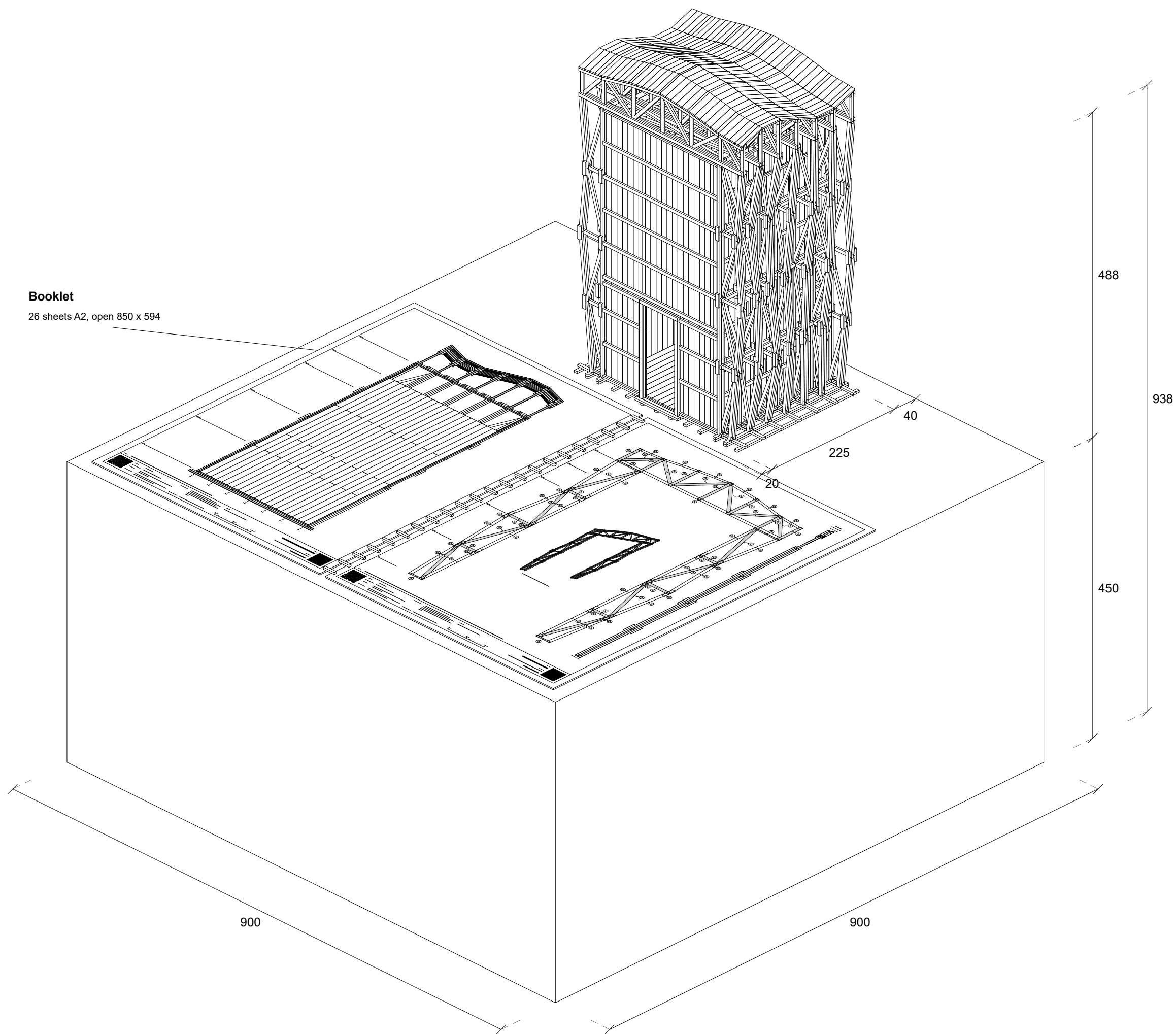
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SCALE Axonometric, scale 1:20, units mm

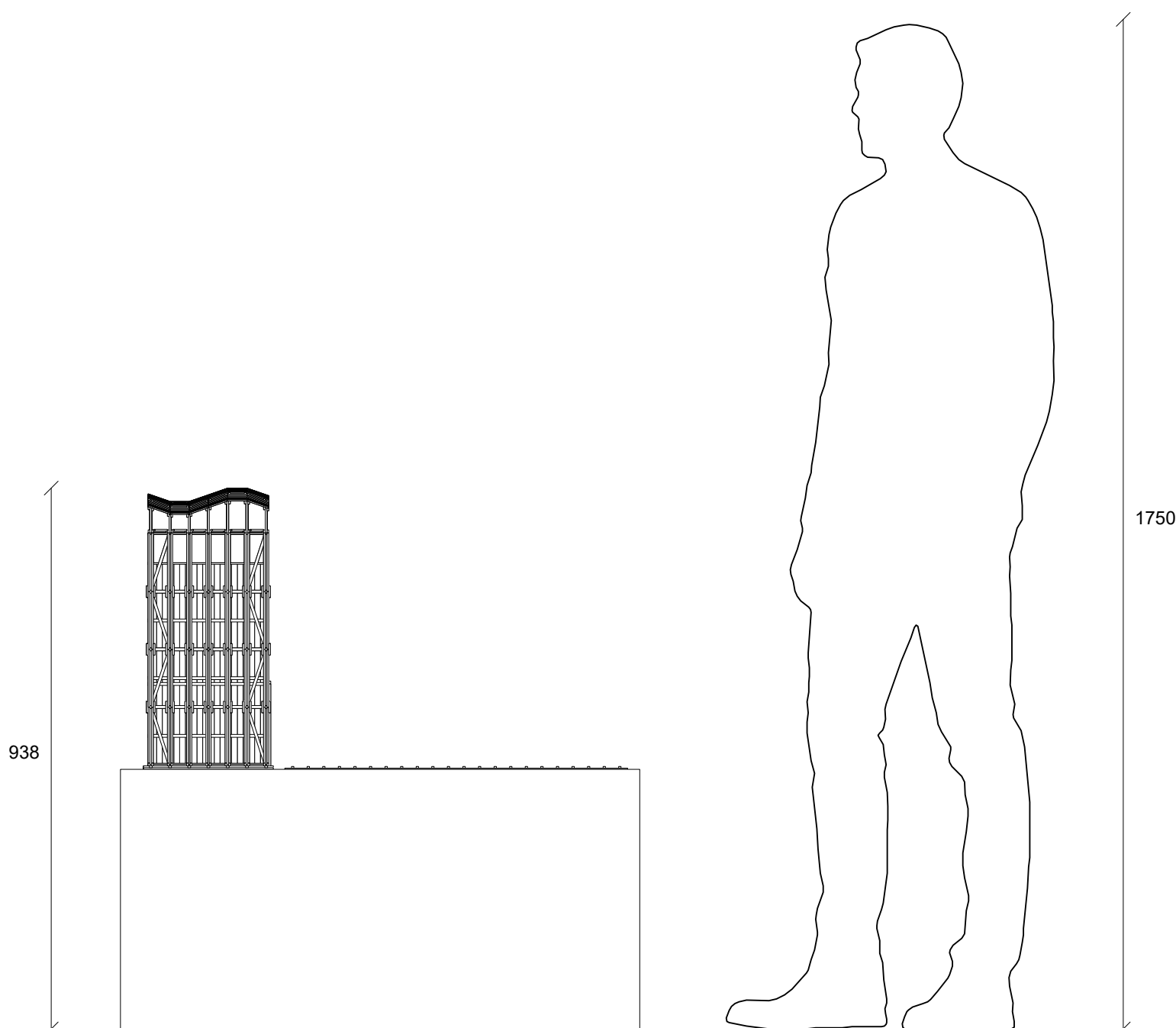


TITLE Axonometric, cladding and fixtures
PROJECT Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET 24



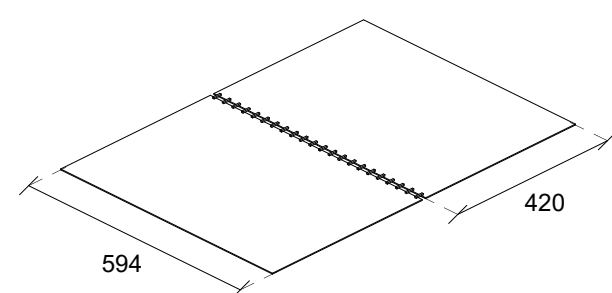
Axonometric 1:5, seen from the door side, with the booklet



Side view 1:10 with a visitor 1.75 m tall

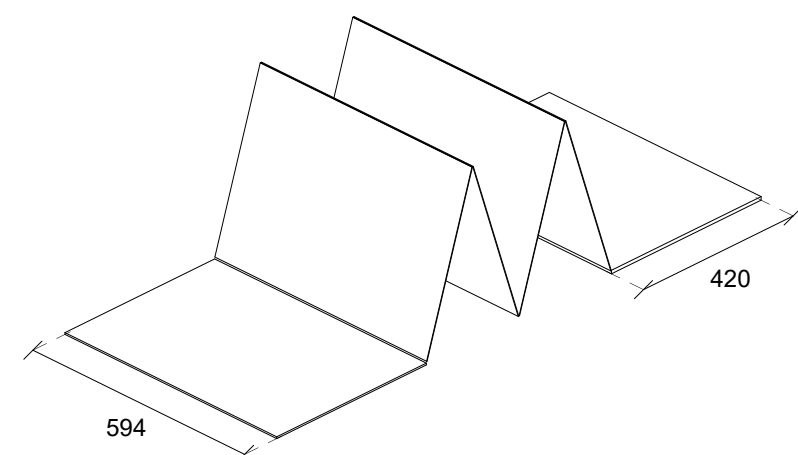
Booklet

26 sheets A2 portrait, spiral bound, open 850 x 594, drawn 1:15



Accordion

26 sheets A2 joined along the long edge, folded 420 x 594, 10.92 m unfolded, drawn 1:15



NOTE Low pedestal 900 x 900 x 450, 938 high with the model (308 x 225 in plan), which stands 40 from the far edge, centred across, its door towards the prints and the visitor. The 26 A2 sheets of this set lie in front of it, open at sheets 12 and 13, as a booklet or as an accordion. The pedestal is drawn as one volume; its boards are not designed yet.



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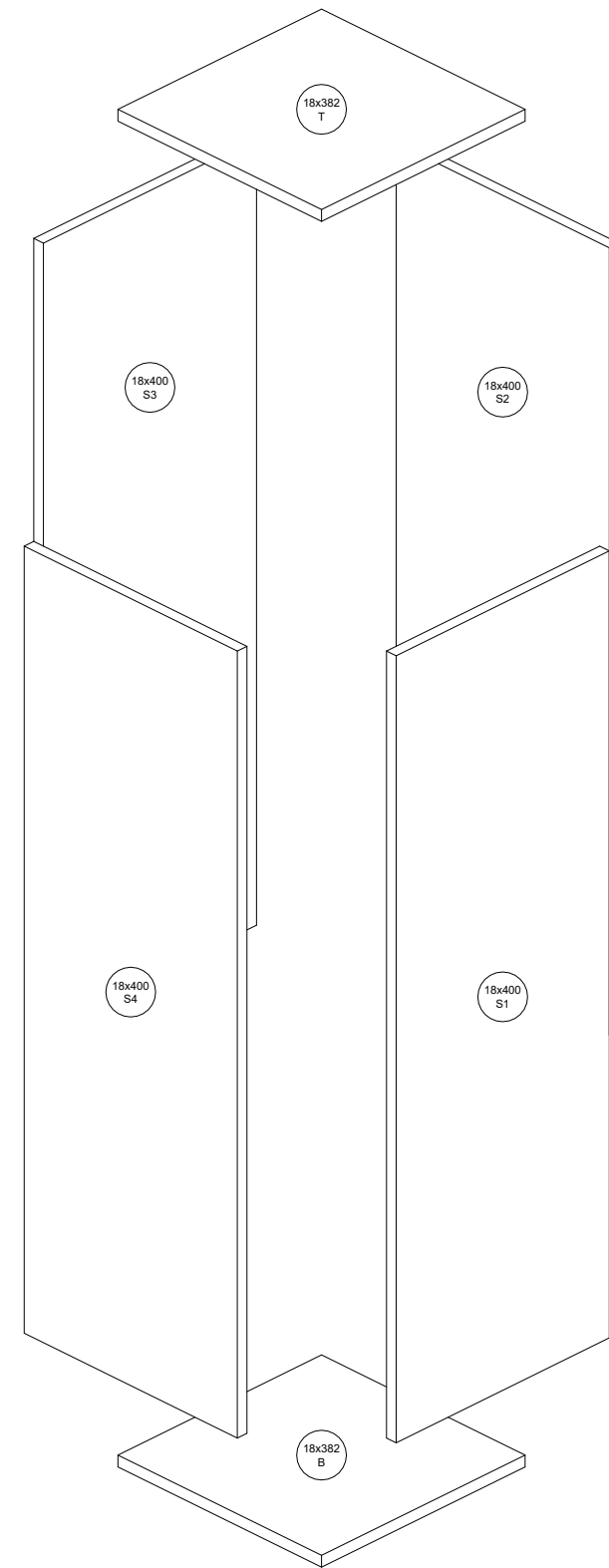
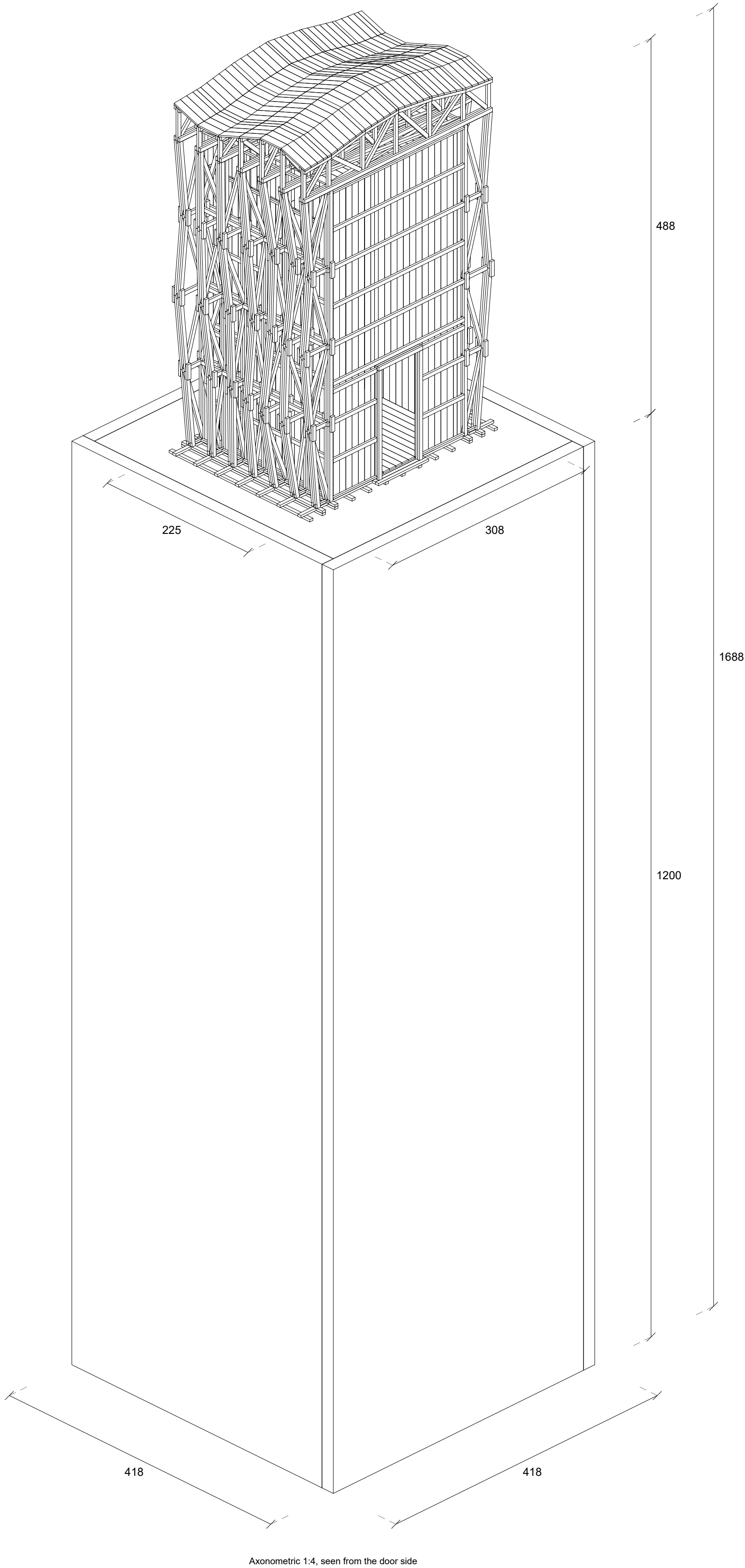
SCALE
Axonometric 1:5, side view 1:10, units mm



TITLE
Low pedestal with model
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol : labs

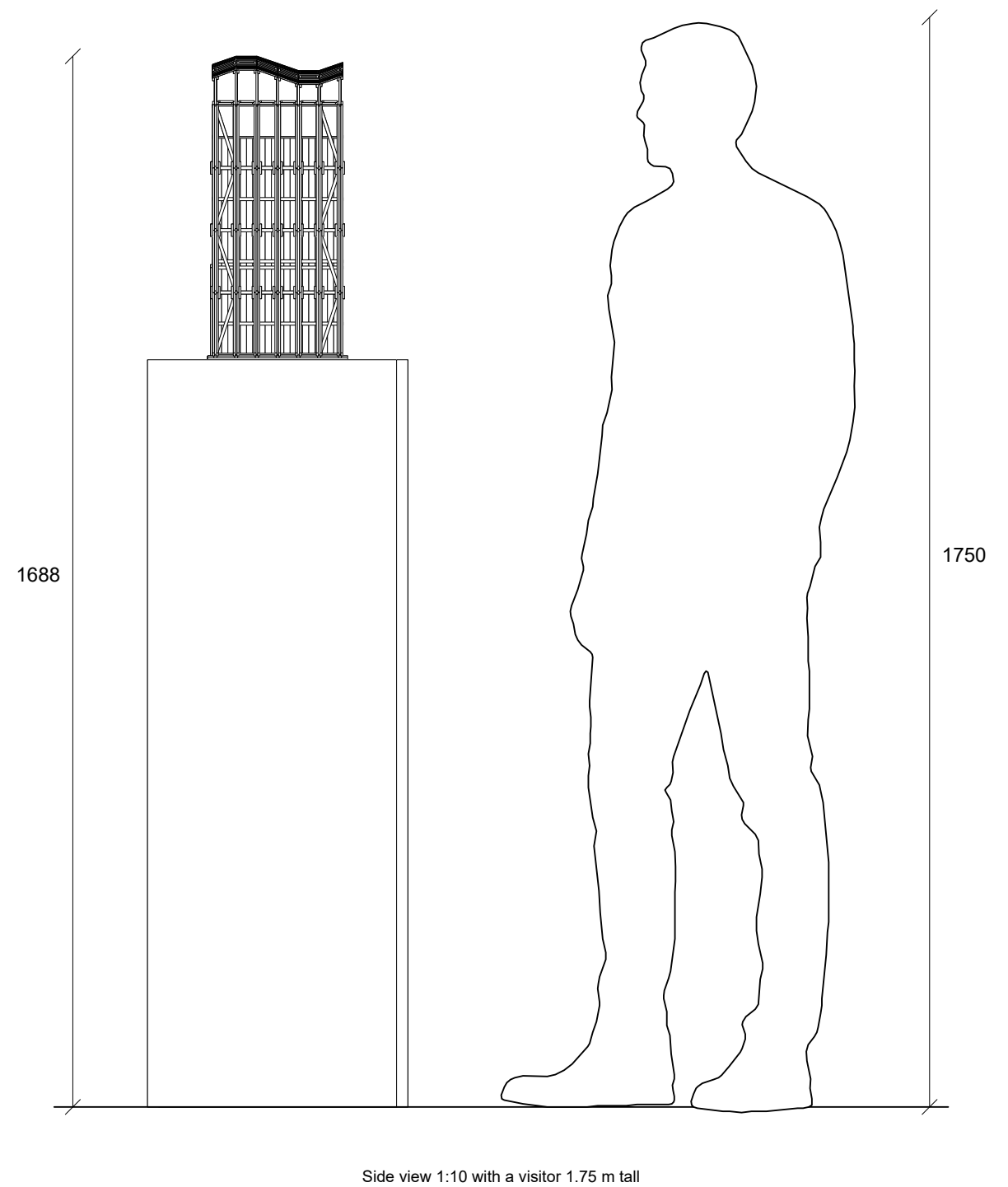
SHEET

25



Boards

S1 to S4	side, 18 x 400 x 1200, whole board	4
T	top, 18 x 382 x 382, cut from board 5	1
B	bottom, 18 x 382 x 382, cut from board 5	1



NOTE Pedestal 418 x 418 x 1200, with the model 418 x 418 x 1688. Each side laps the edge of the next and is screwed into it; top and bottom panels sit inside the sides, flush with their ends, and are screwed through the sides. Circles: board section and board code. The bottom panel carries the ballast.



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SCALE
Axonomic 1:4, others 1:10, units mm



TITLE
Pedestal with model
PROJECT
Experiment 16 - Expressive Structure, GPT-6 v02
protocol:labs

SHEET

26